

Design of an INT4 Dot Product Accelerator for CNN

Full-Custom Design and PPA Analysis

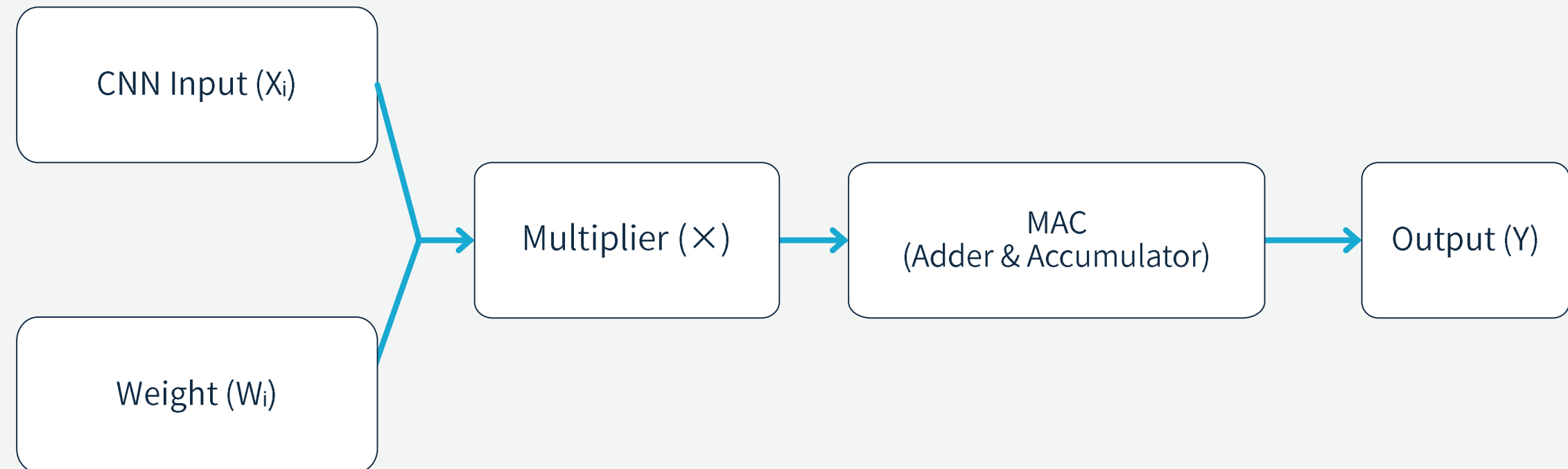
조장 전제현

조원 곽대현 · 김규민 · 박재호 · 이준민

이론

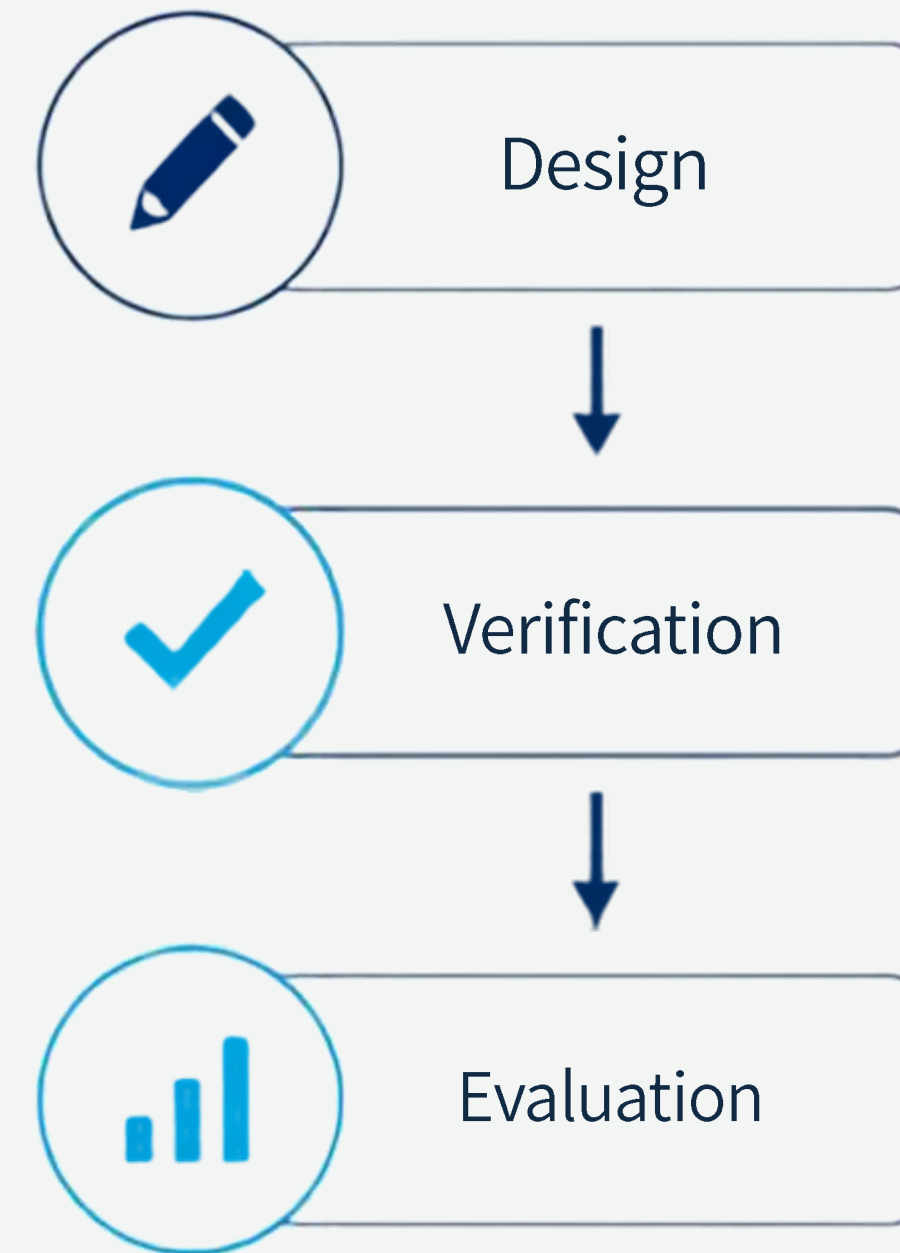
Motivation

- CNN 연산의 대부분은 multiply-accumulate로 구성
- Dot product가 convolution 처리량을 결정
- INT4 저정밀 연산은 computation cost를 줄일 수 있음
- Architecture 선택이 PPA를 좌우

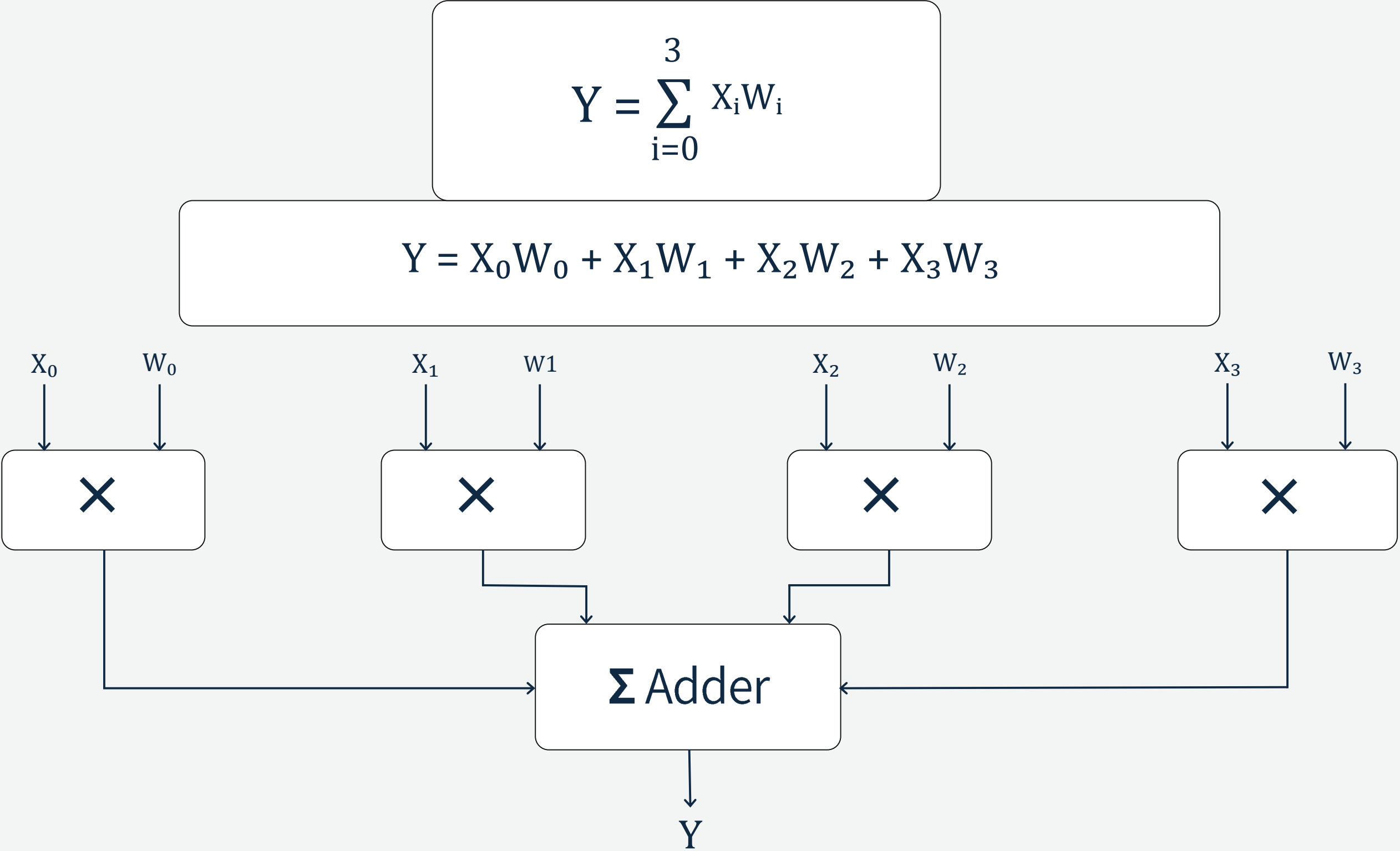


Project Objective

- INT4 dot product accelerator transistor-level / circuit 설계
- Sequential MAC과 Parallel 구조 구현
- Full-custom layout과 DRC 및 LVS 검증
- Post-layout PPA 기반 architecture trade-off 평가



CNN Dot Product Theory



INT4 Quantization

- FP16 data를 4-bit integer로 quantization
- Memory bandwidth와 storage를 최대 4배 절감
- Scaling을 통해 FP16 range를 4-bit integer range로 mapping
- Accumulation overflow 방지를 위한 wider bit-width 설계

FP16 (16-bit)

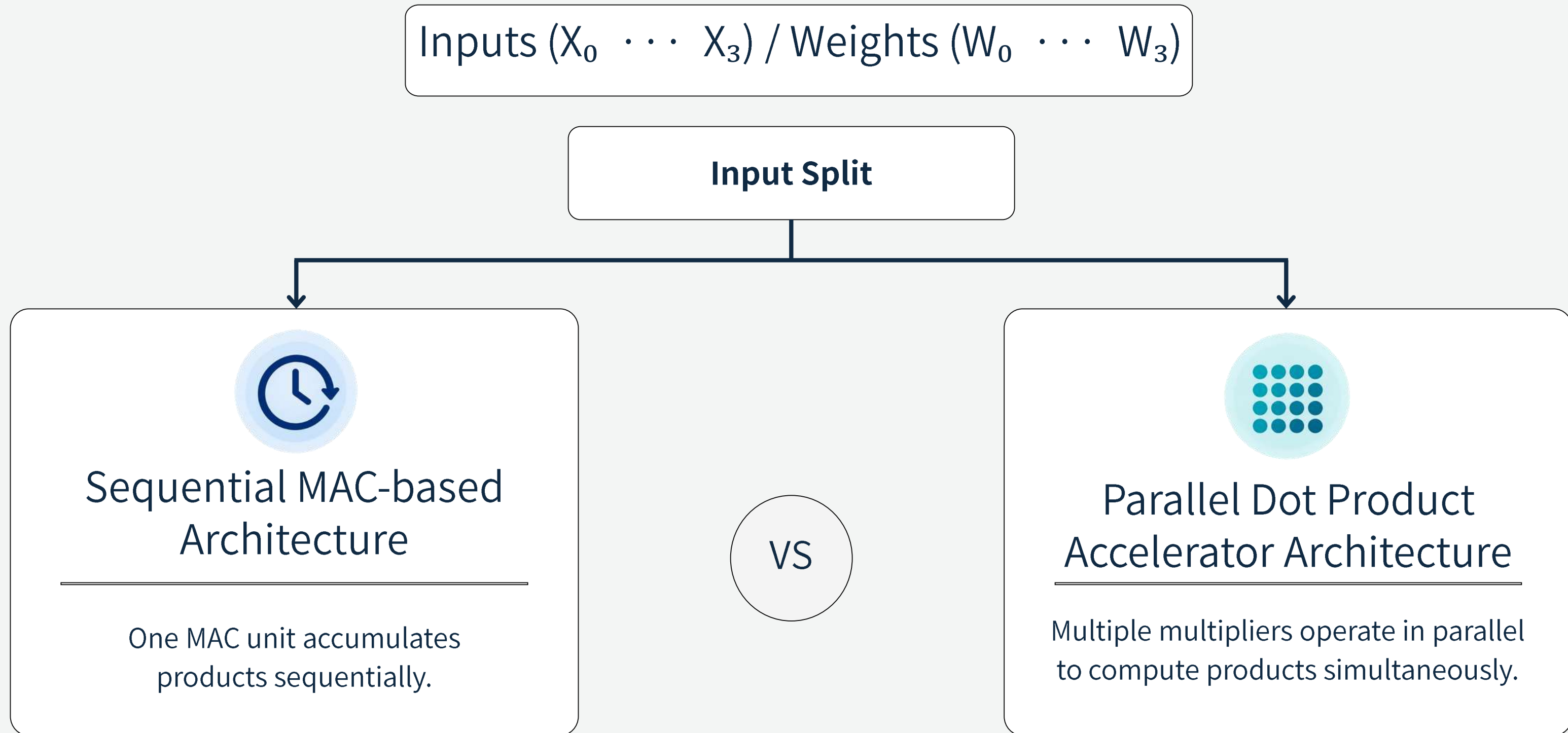


INT4 (4-bit)



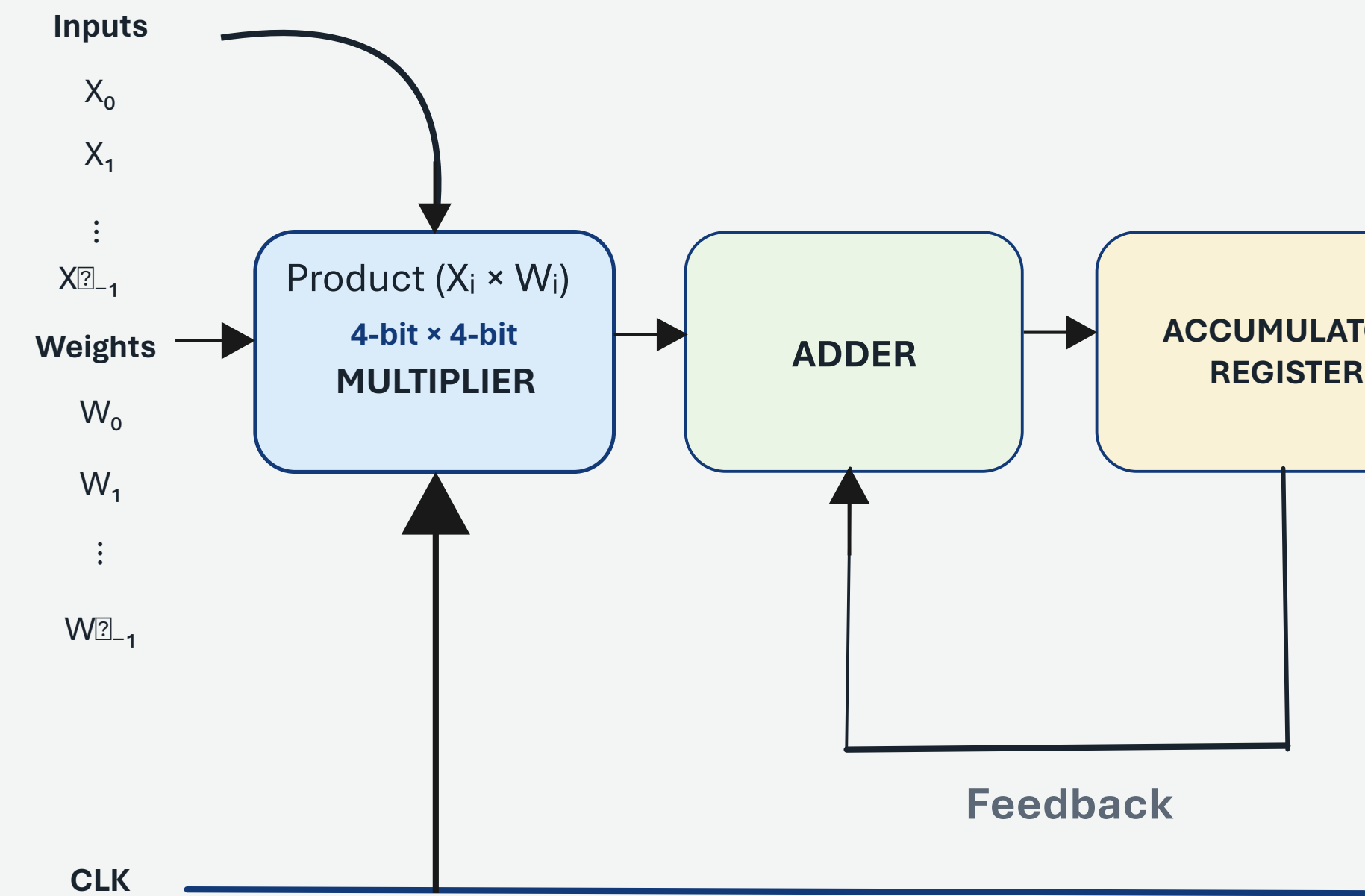
INT4 × INT4 → 8-bit product

Overall Architecture



Sequential MAC Architecture

MAC datapath



Operation

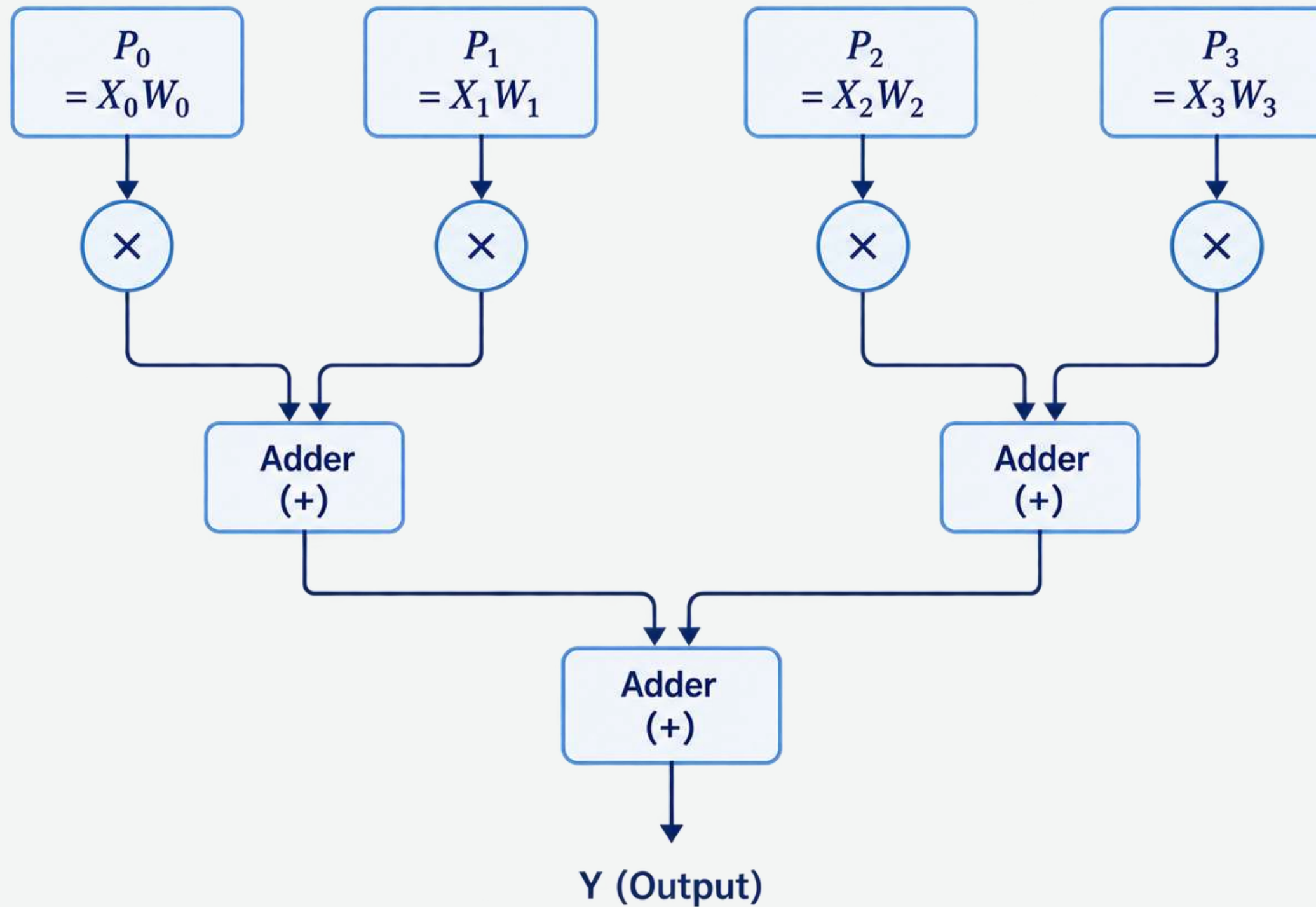
$$ACC_{i+1} = ACC_i + X_i W_i$$
$$Y = \sum X_i W_i$$

Sequential Operation (N = 4)

Cycle	Input	Operation	ACC
0	(X_0, W_0)	$ACC_1 = ACC_0 + X_0 W_0$	$X_0 W_0$
1	(X_1, W_1)	$ACC_2 = ACC_1 + X_1 W_1$	$X_0 W_0 + X_1 W_1$
2	(X_2, W_2)	$ACC_3 = ACC_2 + X_2 W_2$	$X_0 W_0 + X_1 W_1 + X_2 W_2$
3	(X_3, W_3)	$ACC_4 = ACC_3 + X_3 W_3$	$\sum_0^3 X_i W_i$

After N cycles $\rightarrow$ Dot Product Result
$$ACC_n = \sum_{i=0}^{n-1} X_i W_i$$

Parallel Dot Product Accelerator Architecture



Multiplier Design

01

1-bit multiplier = AND gate

02

Multi-bit partial product 생성

03

HA / FA 기반 column reduction

04

Carry propagation으로 final product

Partial Product Generation

0	1	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	1	1	0	0	0
0	0	0	0	1	1	0	0



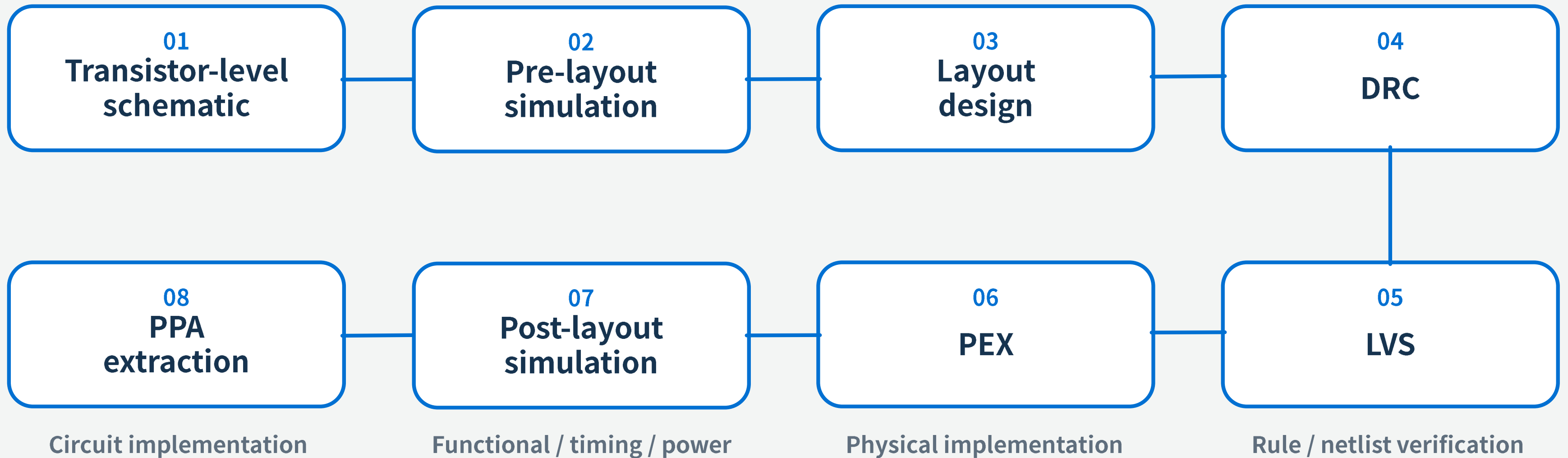
HA / FA Reduction



8-bit Product

P ₇	P ₆	P ₅	P ₄	P ₃	P ₂	P ₁	P ₀
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Full-Custom Design Flow



Transistor-level Design → Physical Verification → Post-layout PPA Evaluation

PPA Evaluation Metrics

01 **AREA**
Layout

Layout 면적 비교

$Area = W \times H$

ACC vs Accelerator

02 **DELAY**
Transient

Input/Output 50% crossing 기준

$t_{pd} = t_{out} - t_{in}$

ADE Transient Simulation

03 **POWER**
Calculator

VDD 소비전력의 평균값

$P_{avg} = (1/T) \int VDD \cdot IDD \, dt$

Virtuoso Calculator

04 **THROUGHPUT**
Performance

단위 시간당 처리 가능한 연산량

$Throughput = Operations / Time$

Parallelism ↑ → Throughput ↑

05 **ENERGY / OP**
Efficiency

연산 1회당 소모 에너지

$E_{op} = E_{total} / Nops$

Power × Time 관점의 효율 비교

MEASUREMENT

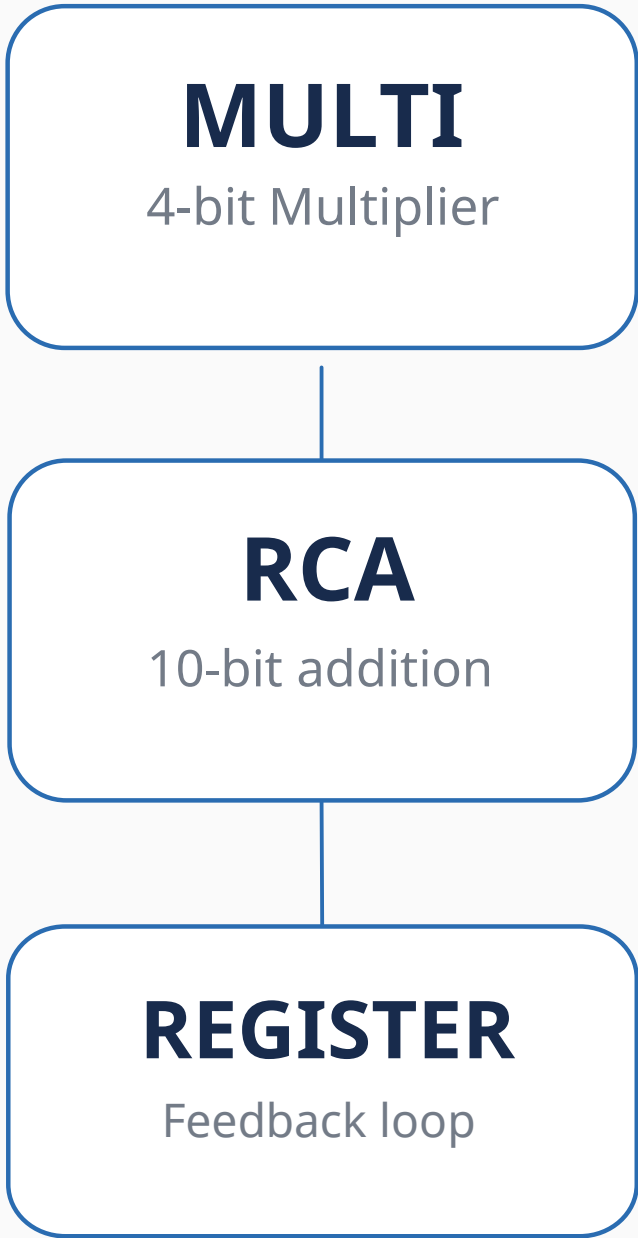
Area → Layout | Delay / Power / Throughput → ADE | Energy/Op → Power × Execution Time

설계 과정

MAC & Dot Product Accelerator Summary

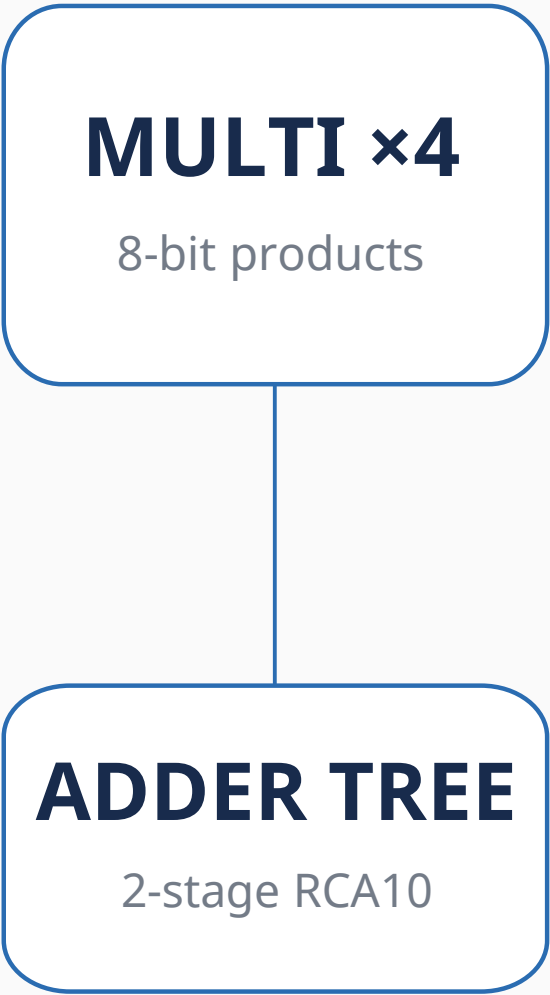
01 MAC STRUCTURE

Flow



02 DOT PRODUCT ACCELEATOR STRUCTURE

Flow



Multiplier Design

01

Partial Product 생성

4-bit 입력에 대한 1-bit AND 기반 partial product

02

HA / FA 기반 column reduction

Partial product를 HA / FA로 정리

03

Carry propagation

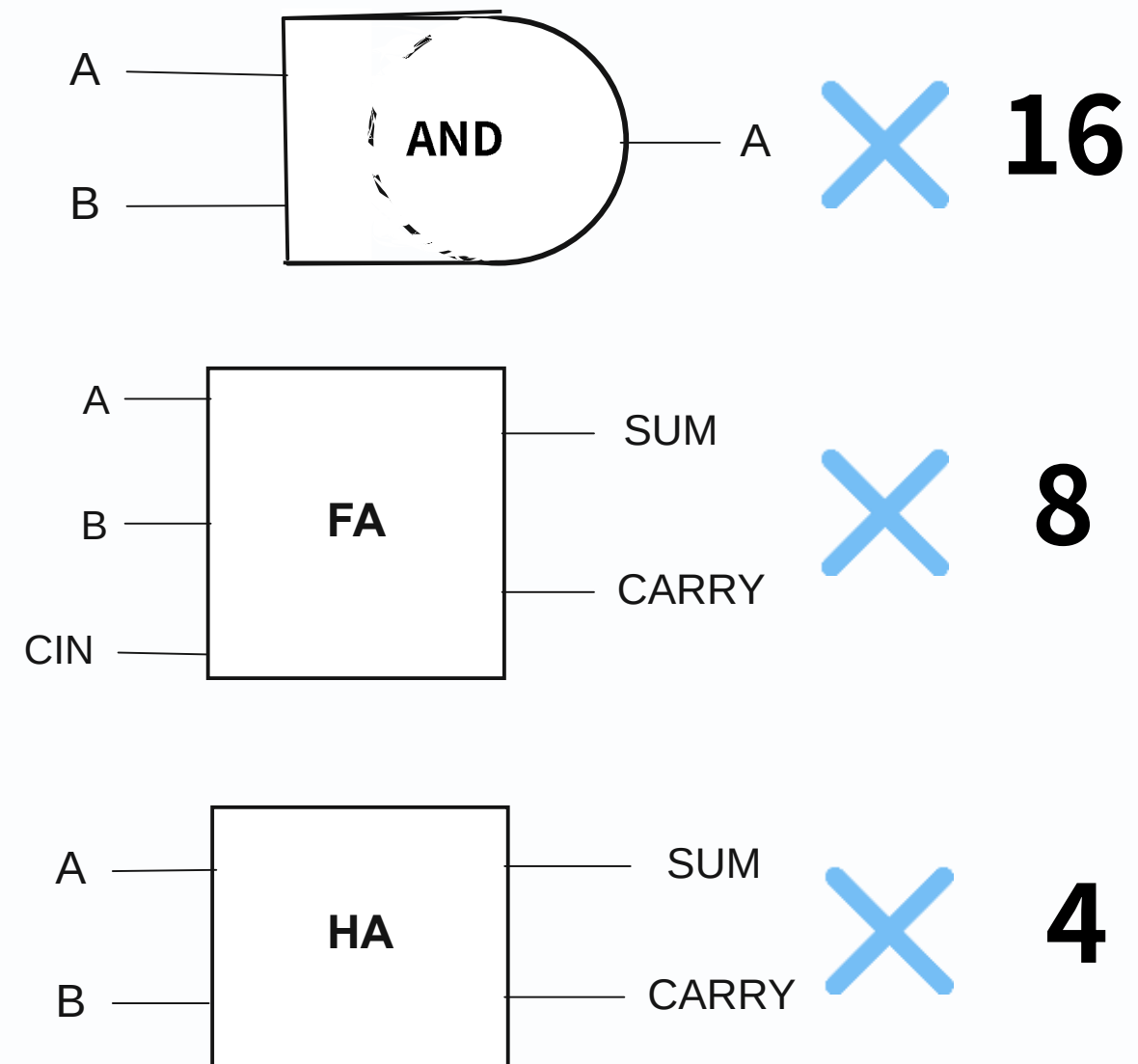
Carry를 전달하여 최종 Product 생성

04

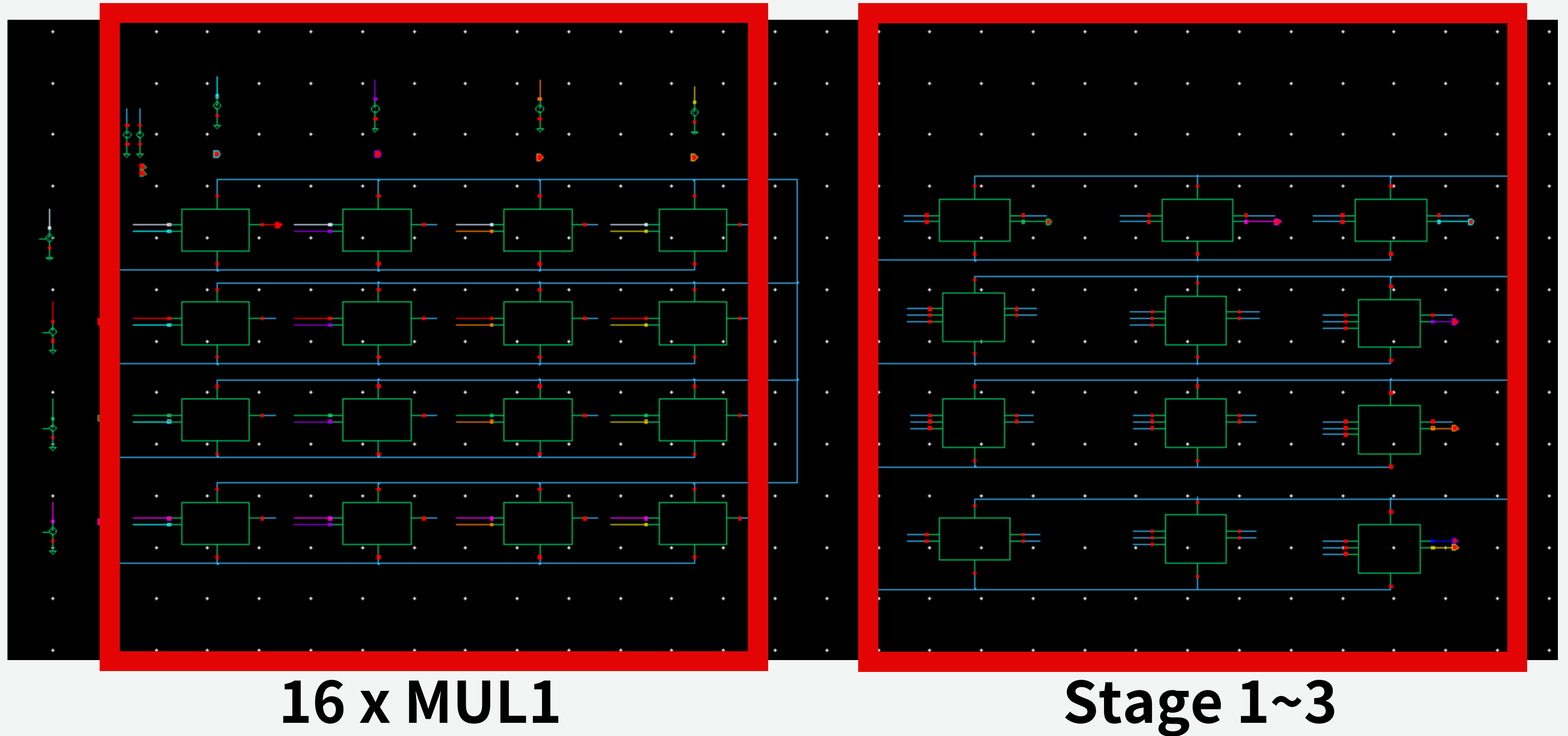
4-bit $\times$ 4-bit $\rightarrow$ 8-bit

P[7:0] 출력

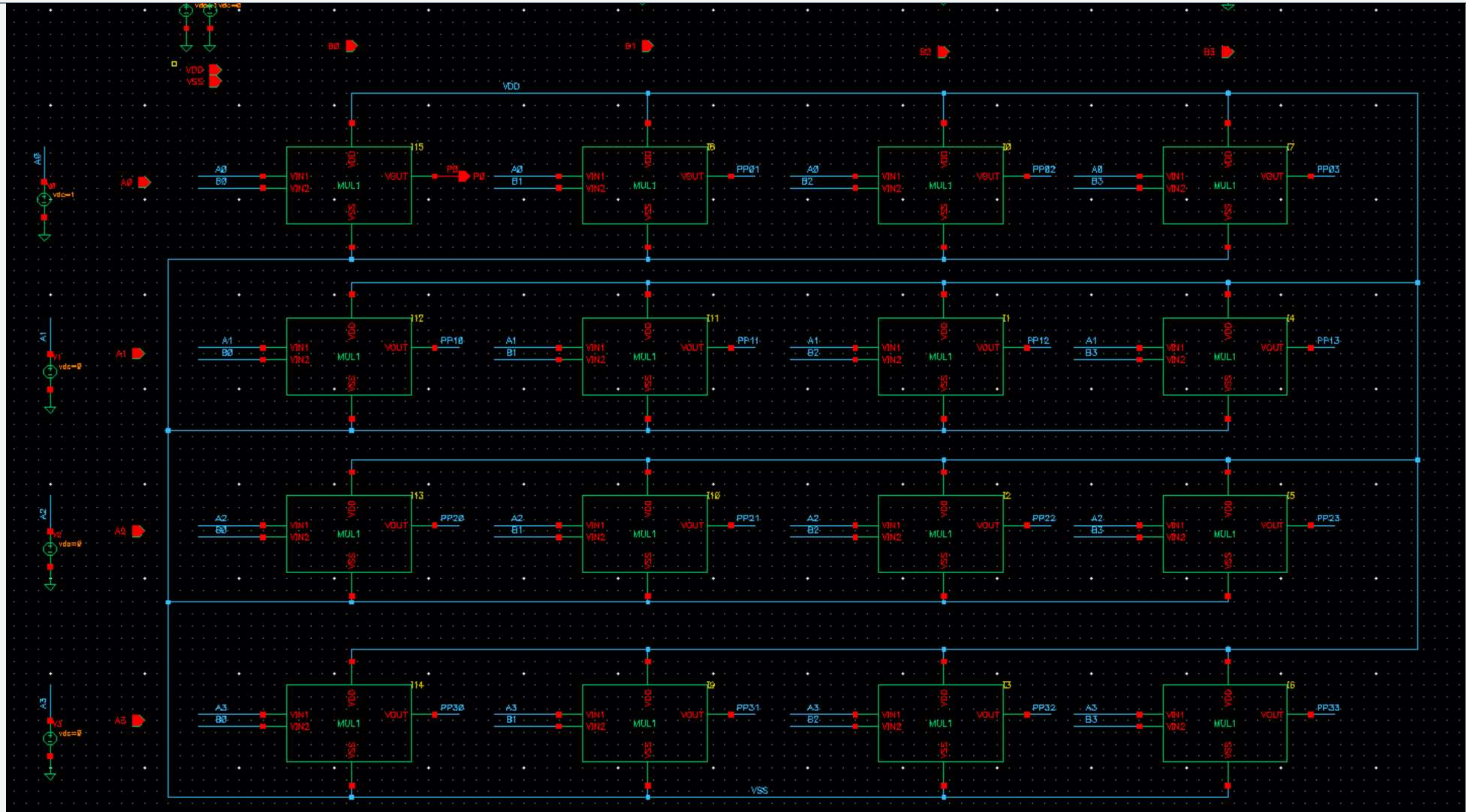
Component



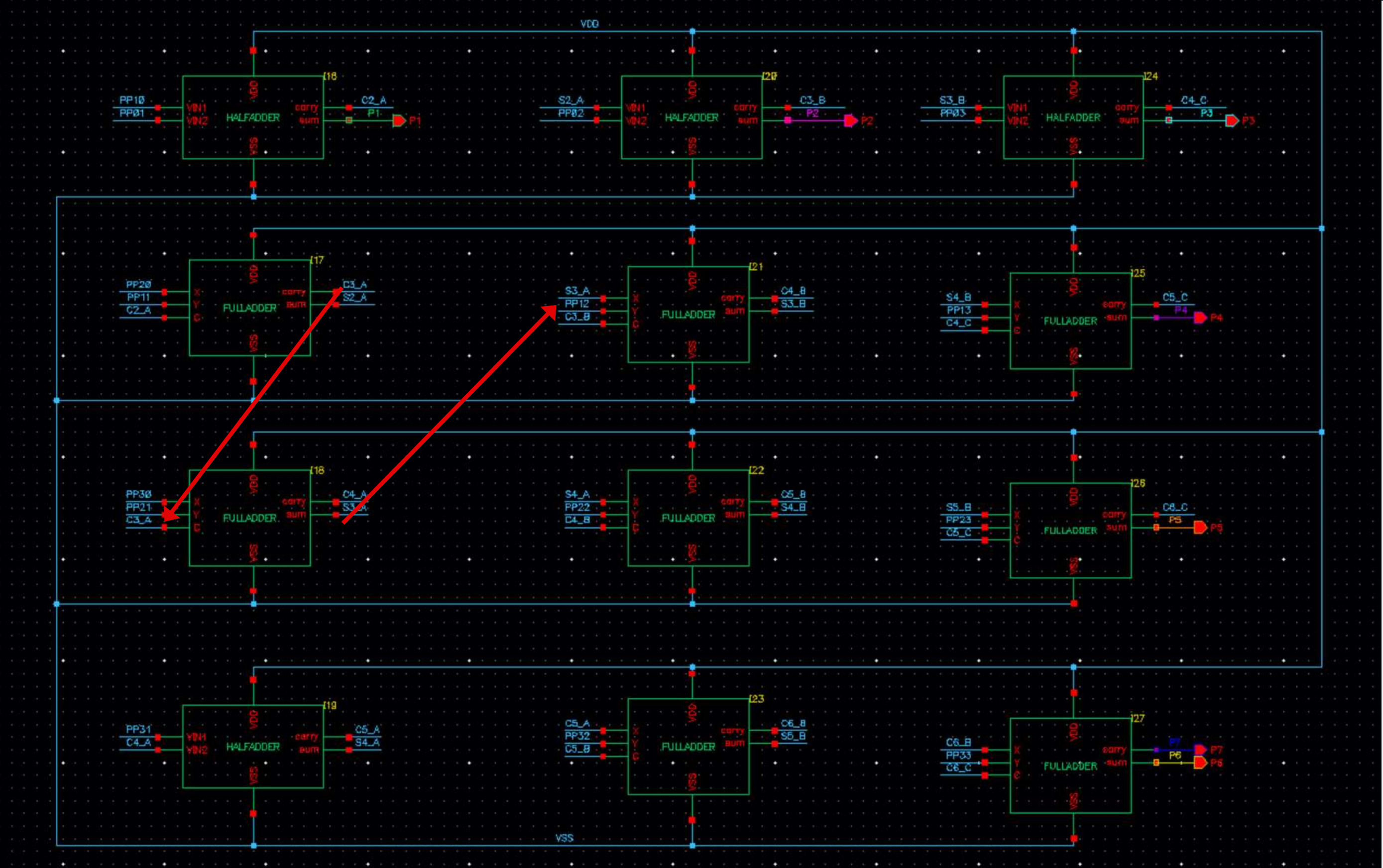
Multiplier Schematic



Schematic (16 x MUL1)



Schematic (Stage 1~3)



RCA10 Design

01

1-bit Adder 구성

HA / FA를 이용한 기본 가산기 구성

02

10-bit RCA 구성

10개의 Full Adder를 연결

03

Ripple Carry

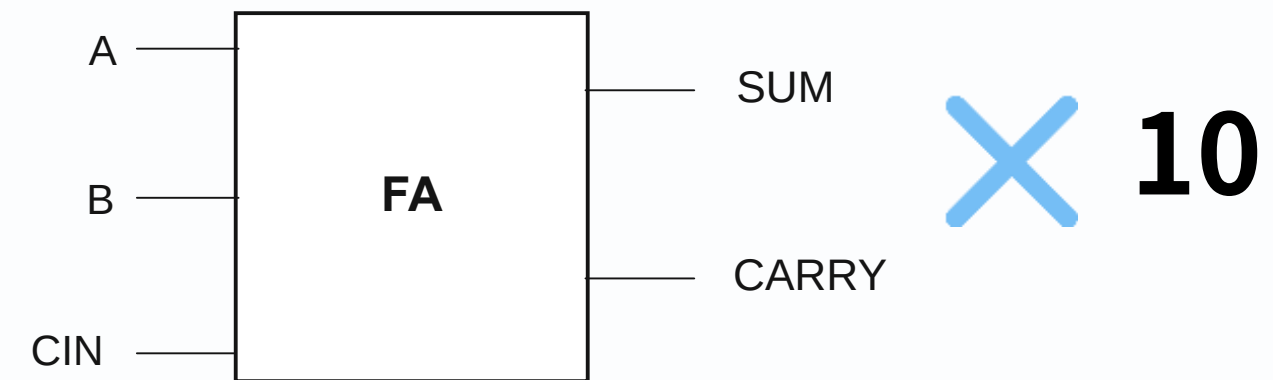
각 stage의 Carry를 다음 stage로 전달

04

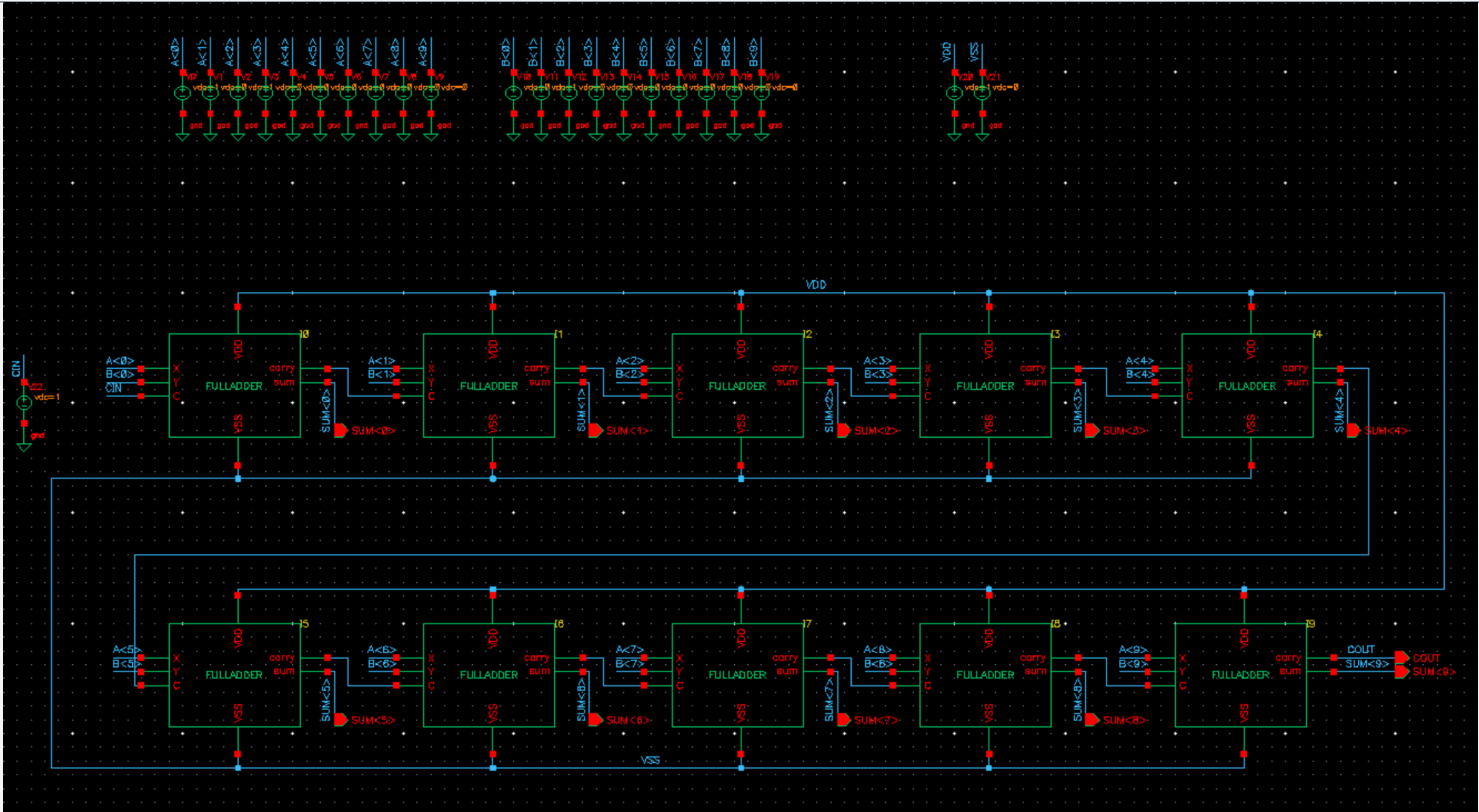
SUM[9:0] 출력

MAC accumulation 및 Adder Tree에 활용

Component



RCA10 Schematic



Register Design

01

Register 구조 설계

10-bit 누산값 저장을 위한 Register 구성

02

Schematic 구현

공통 CLK / Reset 기반 DFF 연결

03

Layout 구현

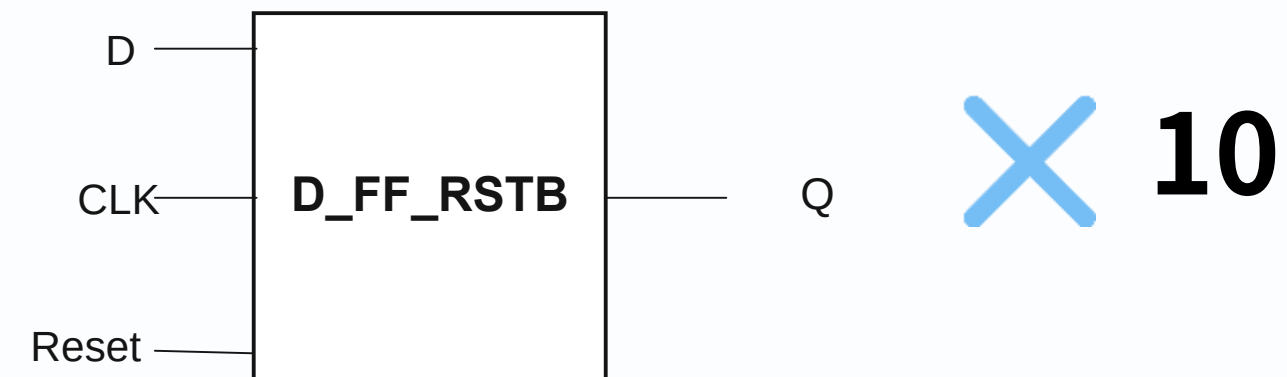
10개의 DFF 배치 및 Clock / Data Routing

04

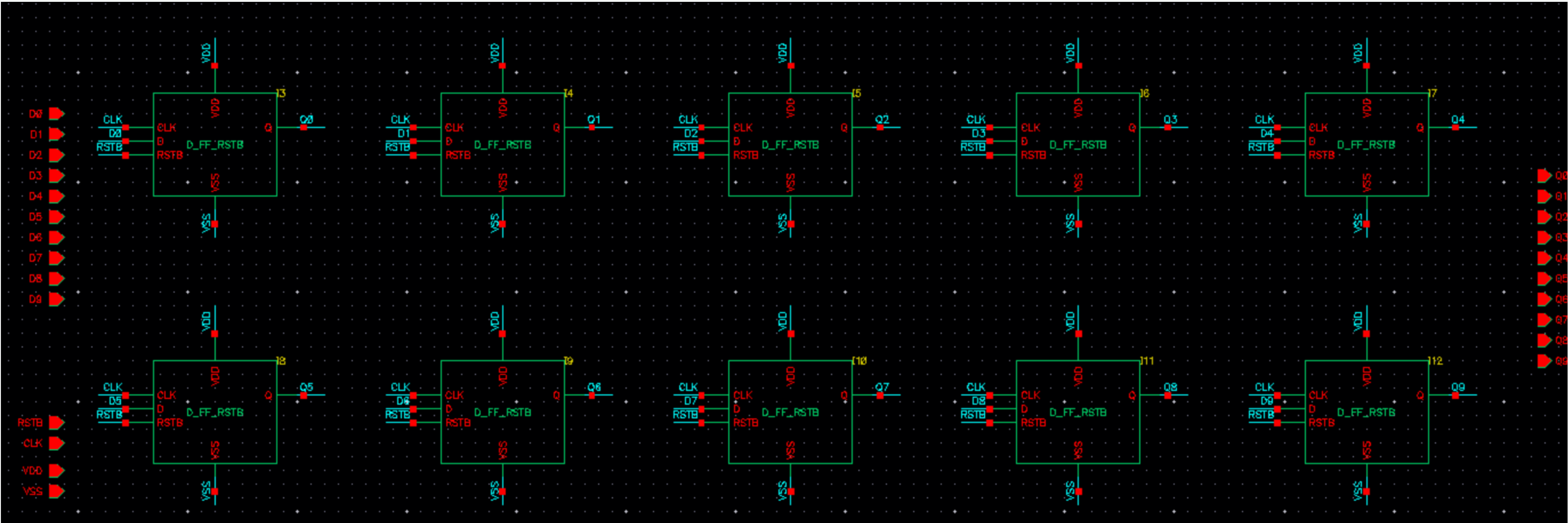
검증

Register 동작 및 Physical Verification

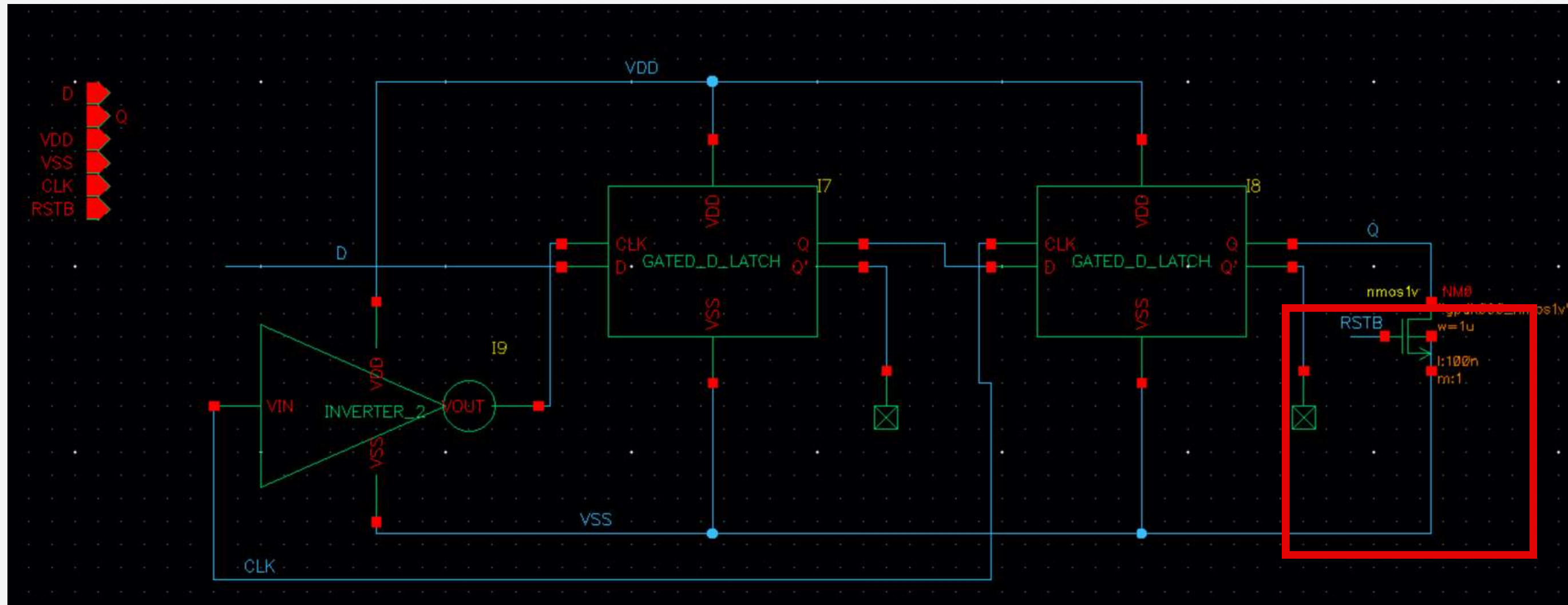
Component



Register Schematic



D-FF 초기화 설계: NMOS 기반 RESET 구조 적용



01. 초기 상태 설정

- Register의 초기 출력값 0으로 설정 → 시뮬레이션 및 연산 안정성 확보

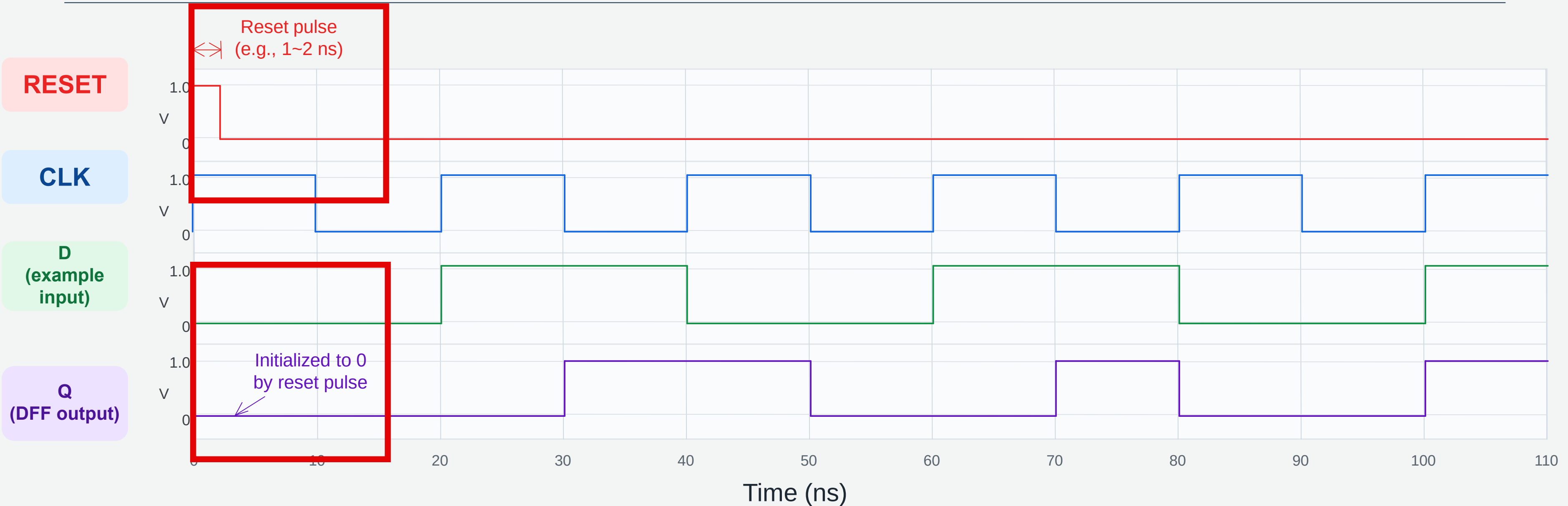
02. NMOS 추가

- D-FF 내 NMOS를 추가하여 VSS(0V)와 연결

03. Reset 동작

- RSTB 해제 시 NMOS가 OFF 되어 D-FF의 정상적인 데이터 저장 동작 수행

D-FF 초기화 설계: NMOS 기반 RESET 구조 적용



- RESET is asserted to 1 V for a very short time before the first clock rising edge.
- After the pulse, RESET remains low (0 V).
- The DFF output Q is initialized to 0 and then follows the D input on each rising edge of CLK.

Simulation Conditions (example)

- VDD = 1.0 V
- Reset pulse width = 1 ~ 2 ns
- Clock period = 20 ns (50 MHz)
- D: example pattern

MAC Design

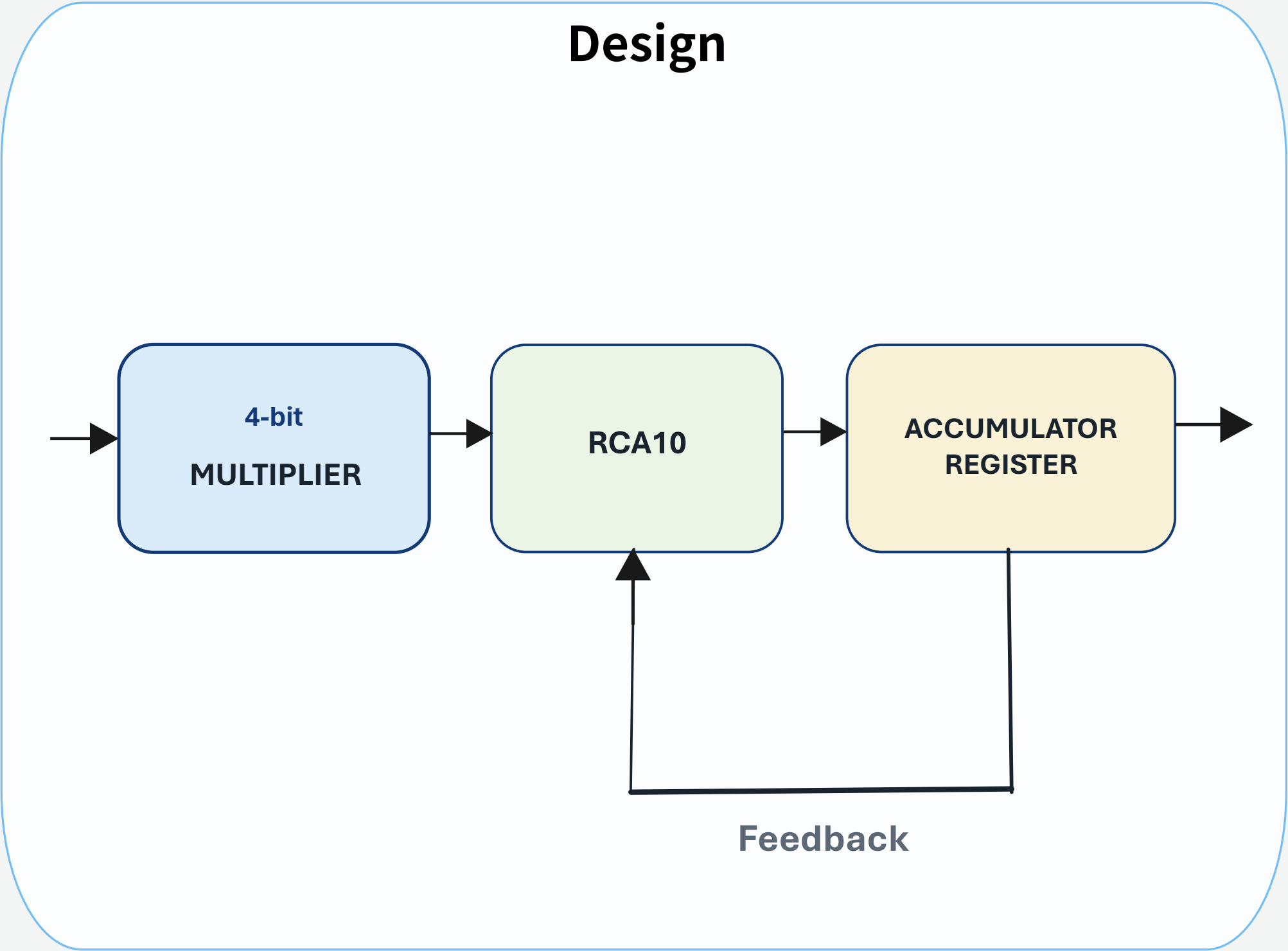
- 01

MAC 구조 설계
4-bit Multiplier 및 Adder 기반 MAC 구조 구성
- 02

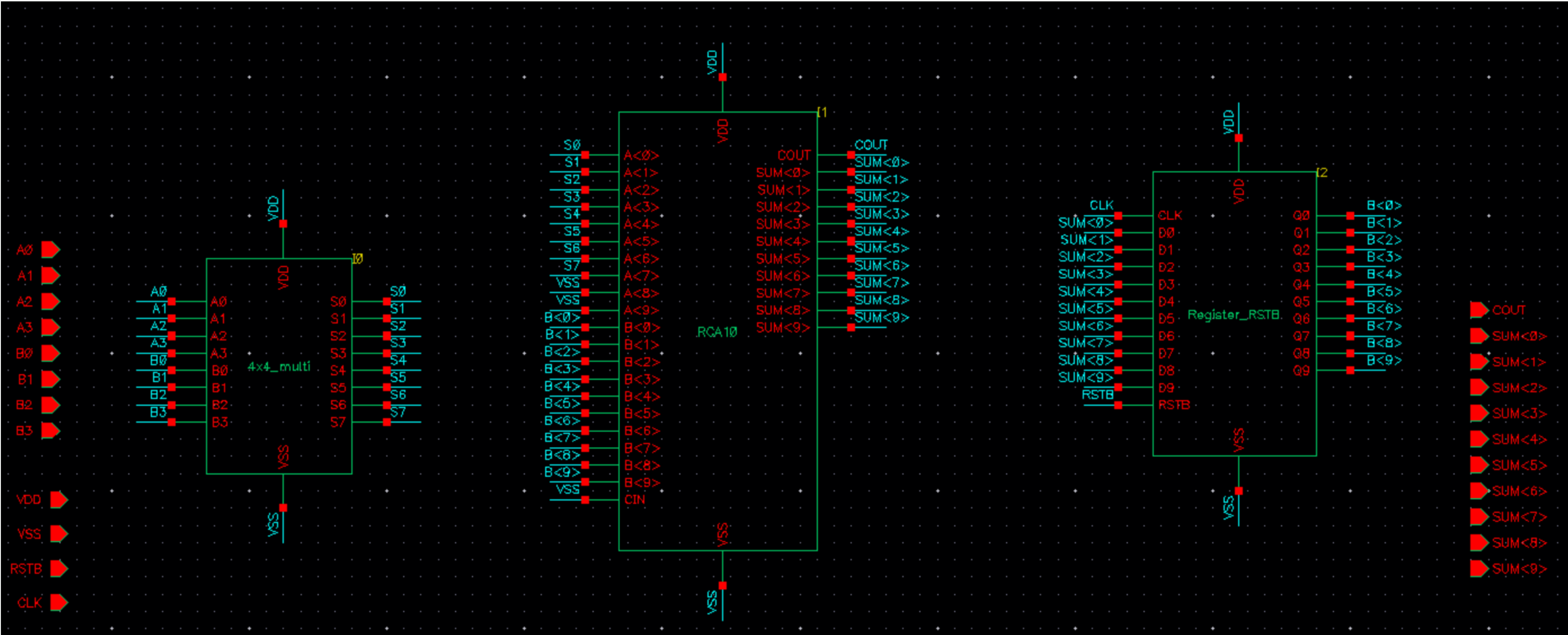
Schematic 구현
Multiplier, RCA 및 Register 연결
- 03

Layout 구현
MAC 회로의 물리적 배치 및 Routing
- 04

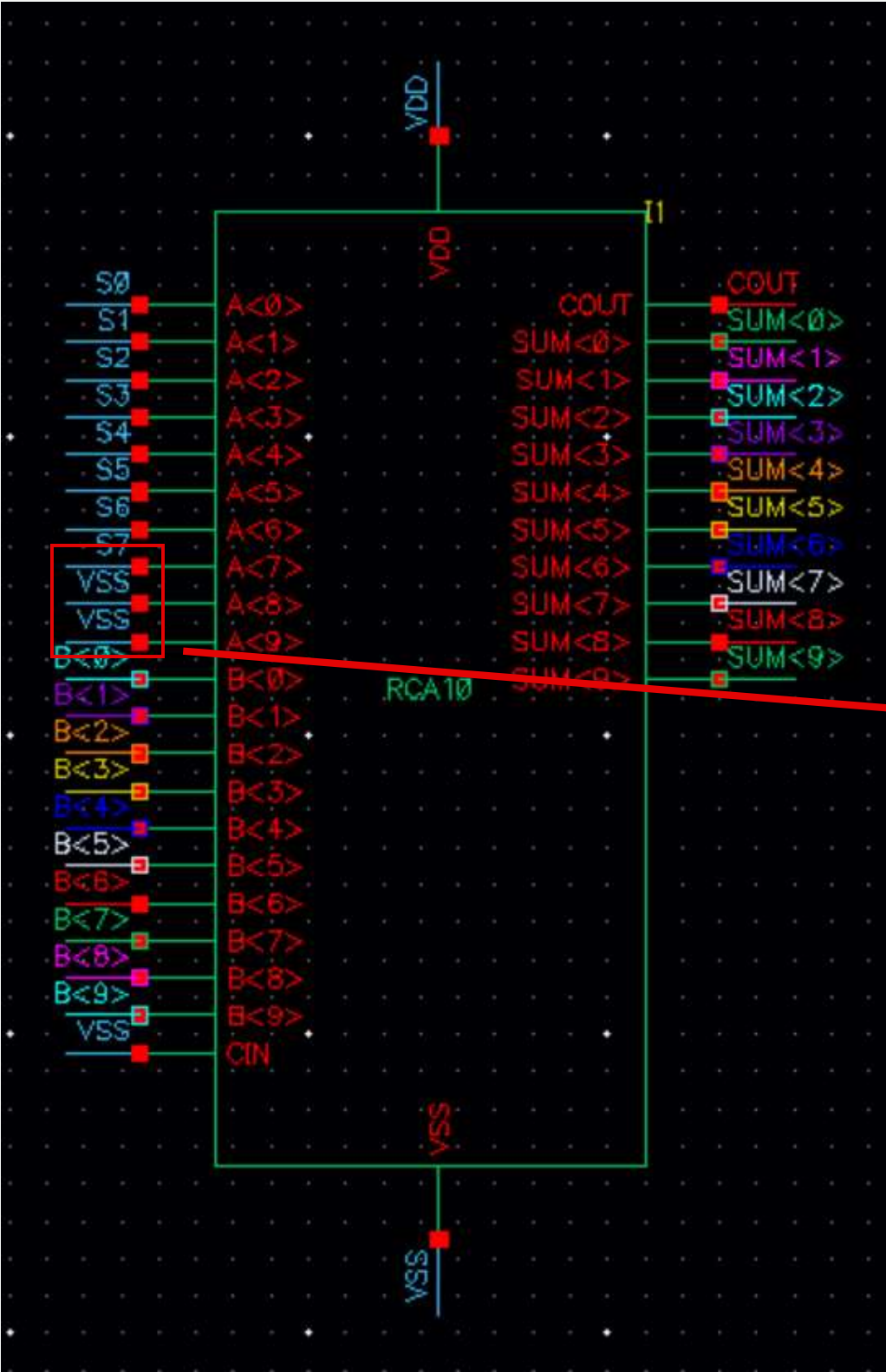
검증
Layout 검증 및 곱셈·누산 결과 확인



MAC Schematic



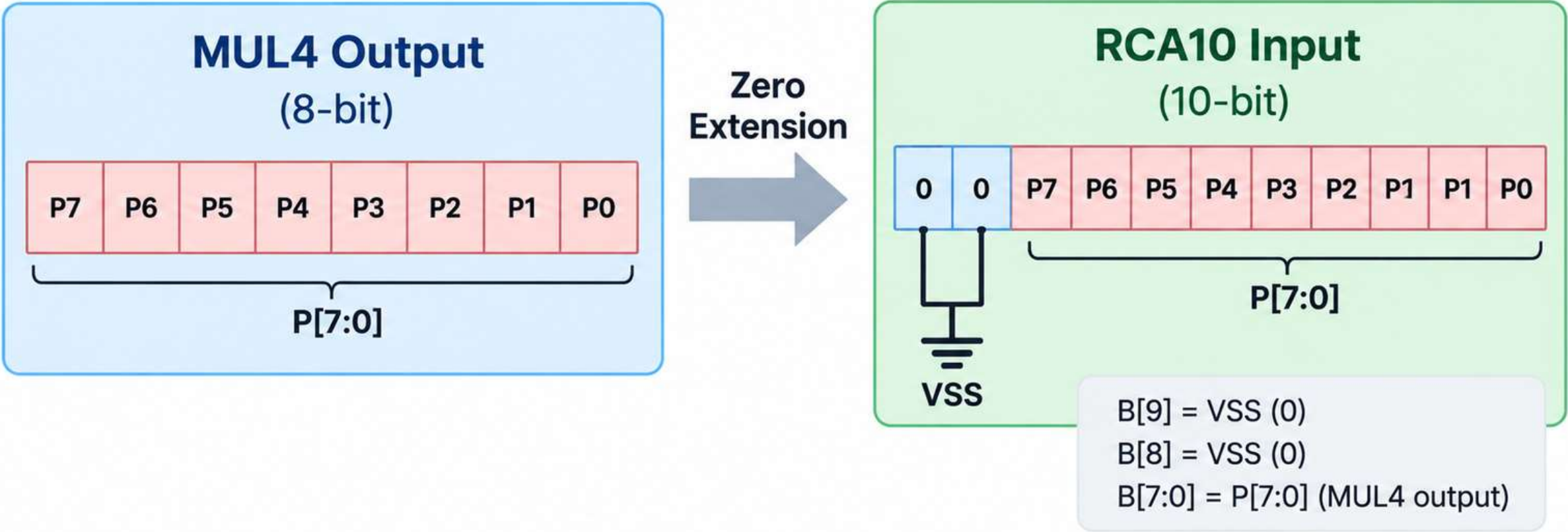
MAC Schematic - Zero Extension



Zero Extension

(Connect MSBs to VSS)

Extend the 8-bit multiplier output to 10-bit by connecting the two MSBs to VSS (logic 0).



Dot product Accelerator Design

01

Dot Product Accelerator 구조 설계

4개의 MUL4와 3개의 RCA10으로 2-stage Adder Tree 구성

02

Schematic 구현

Multiplier와 RCA를 연결하여 4-way Parallel Dot Product 회로 구현

03

Layout 구현

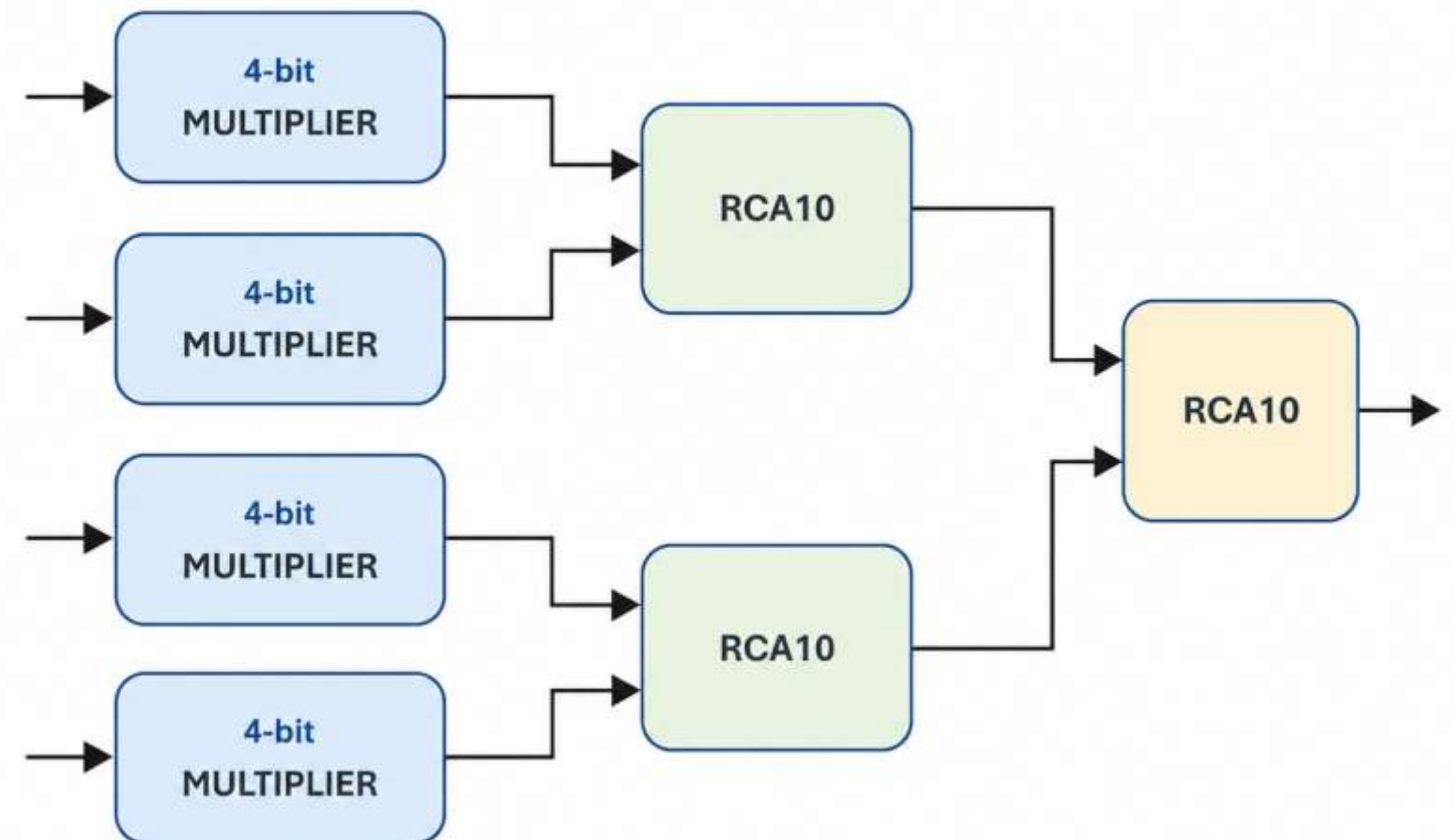
각 연산 블록을 배치하고 bit별 신호 Routing

04

검증

Dot Product 동작 및 Physical Verification

Design



Adder Tree Design

01

4개의 Product 입력

4-bit Multiplier $\times 4$

02

1st Stage

RCA10 $\times 2$: P_0+P_1, P_2+P_3

03

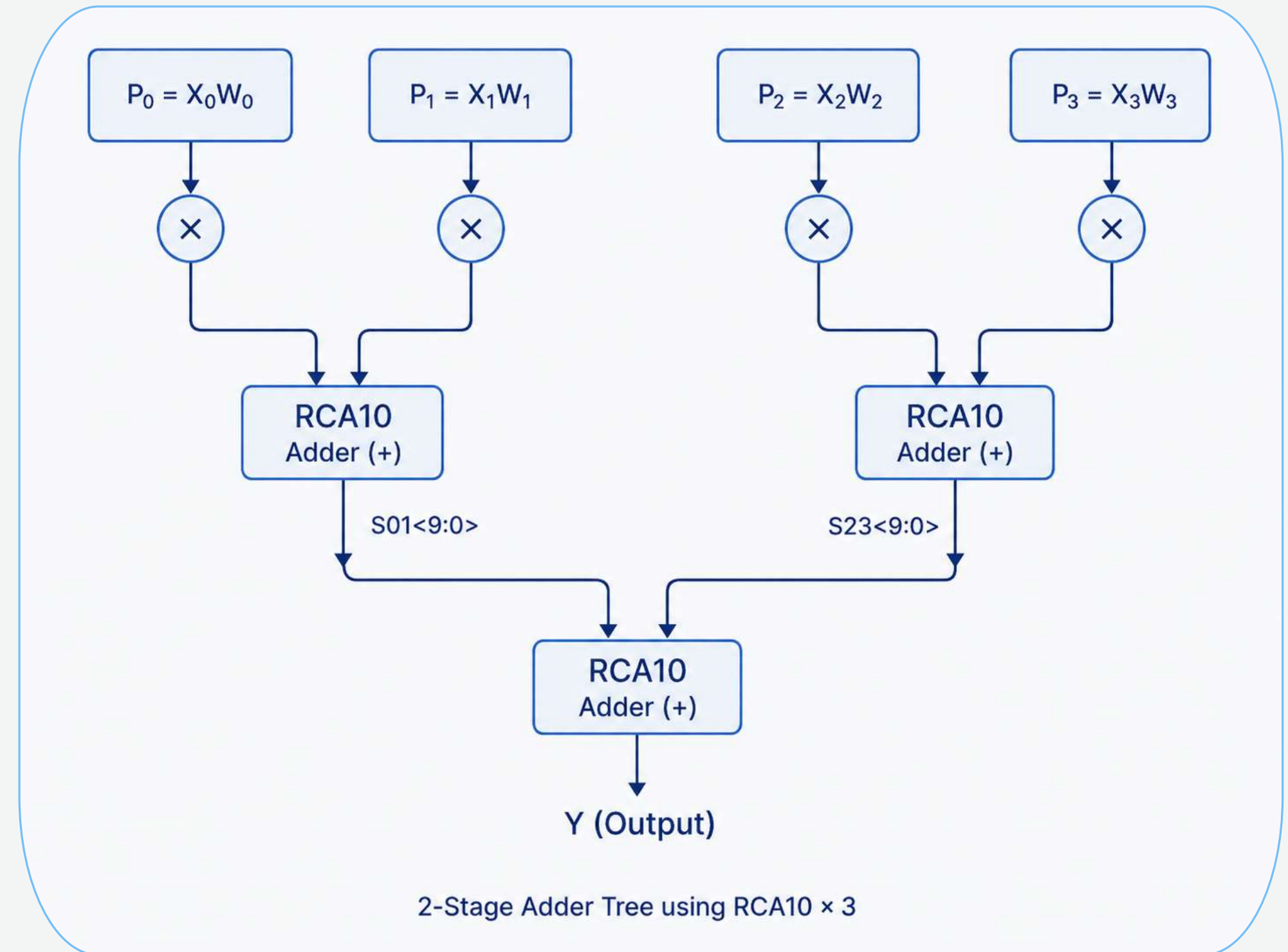
2nd Stage

RCA10 $\times 1$: 두 중간 결과를 최종 합산

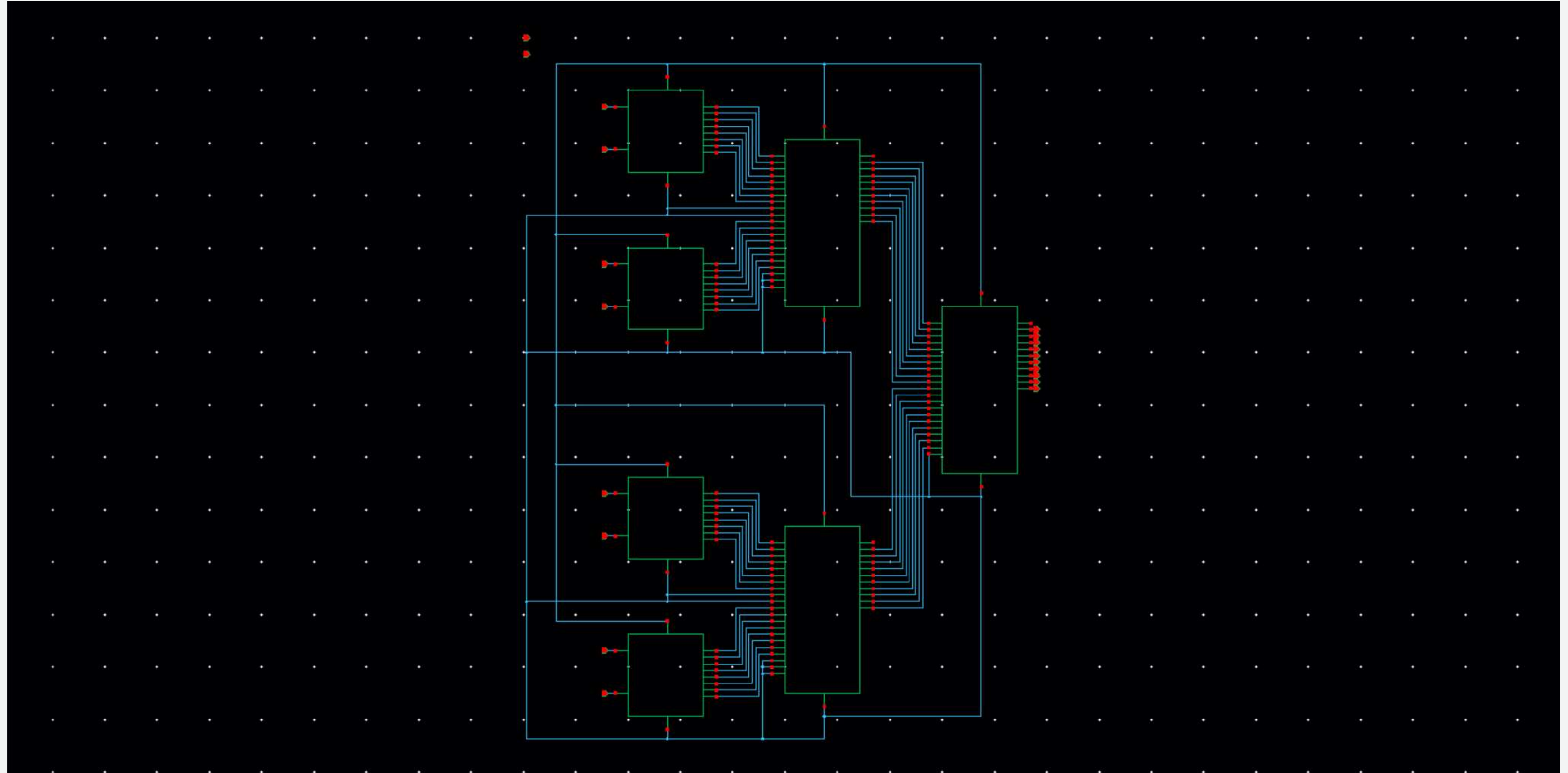
04

Dot Product 출력

8 $\rightarrow$ 9 $\rightarrow$ 10-bit width growth



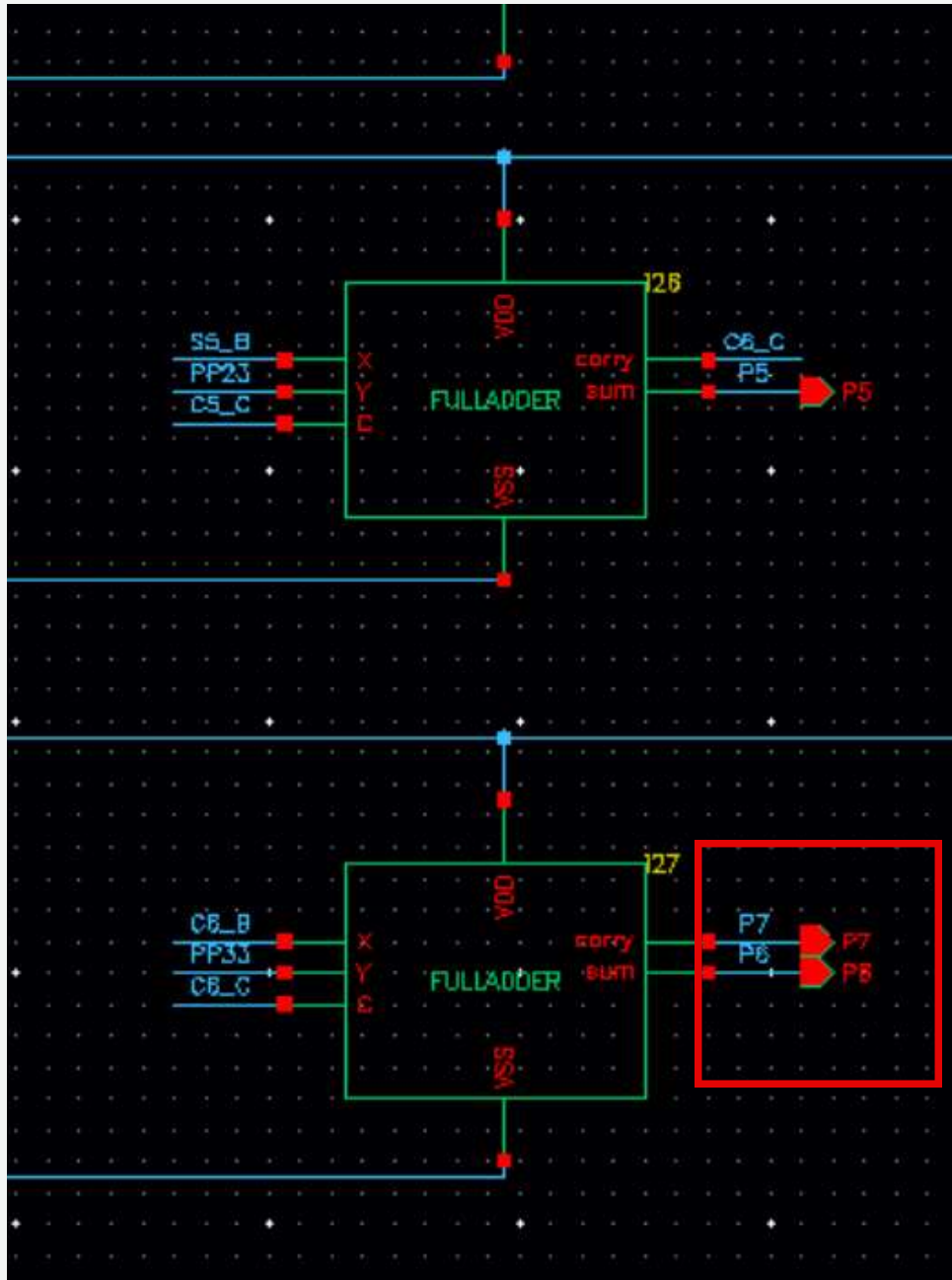
Dot Product Accelerator Schematic



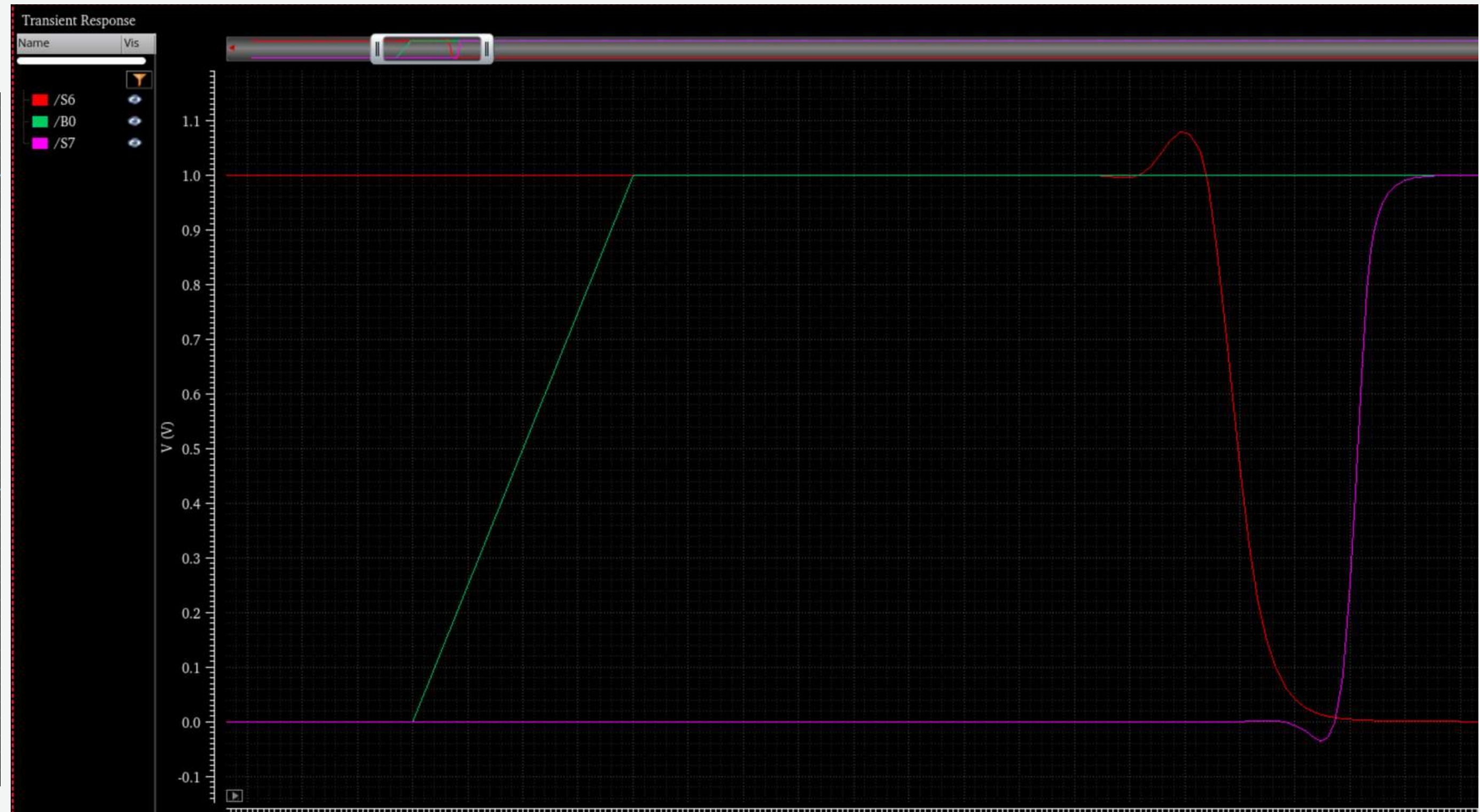
Simulation - Multiplier



Propagation Delay time (Critical path) - Multiplier

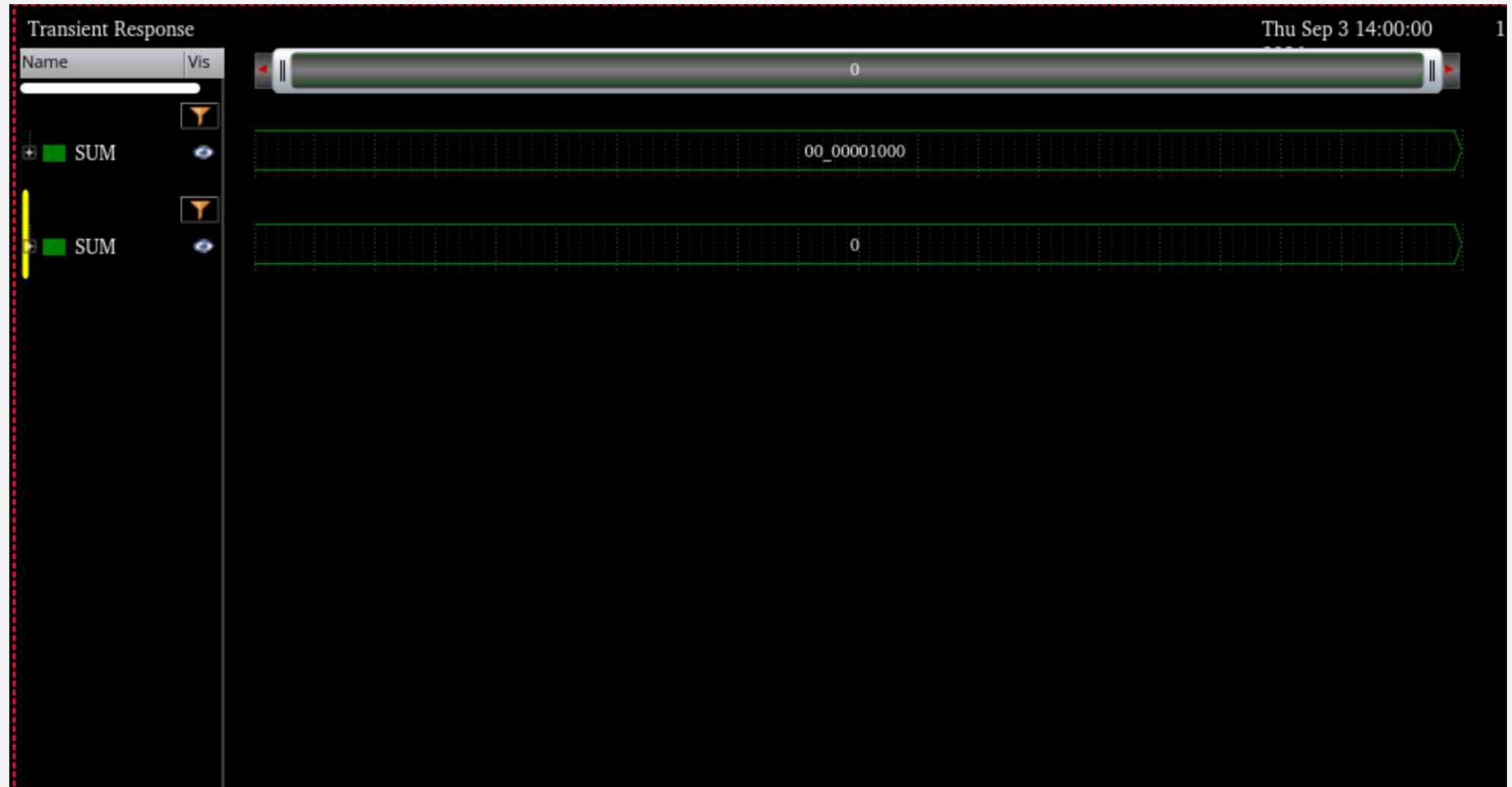


Last stage

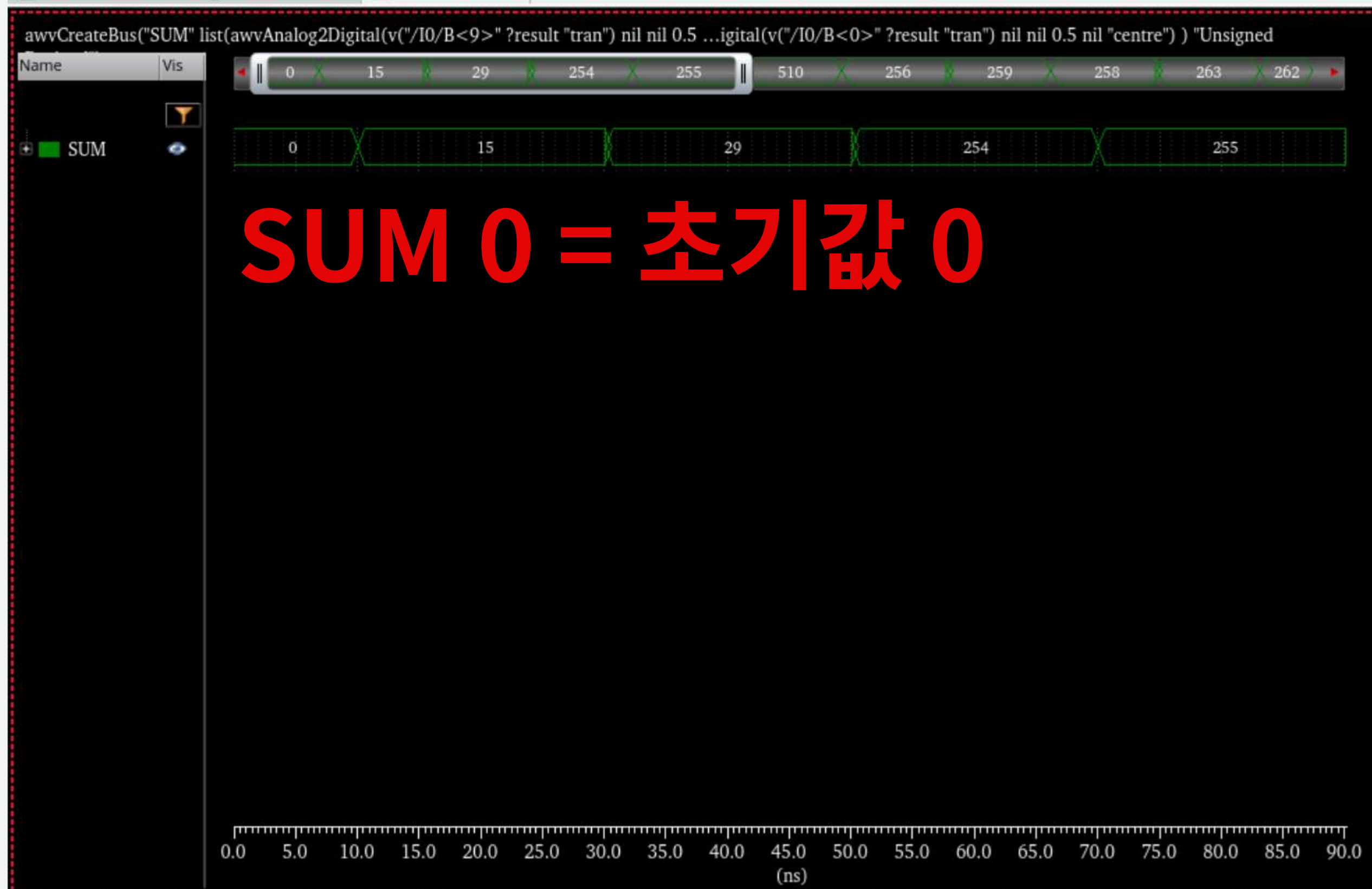


Delay time: 0.385ns

Simulation - RCA

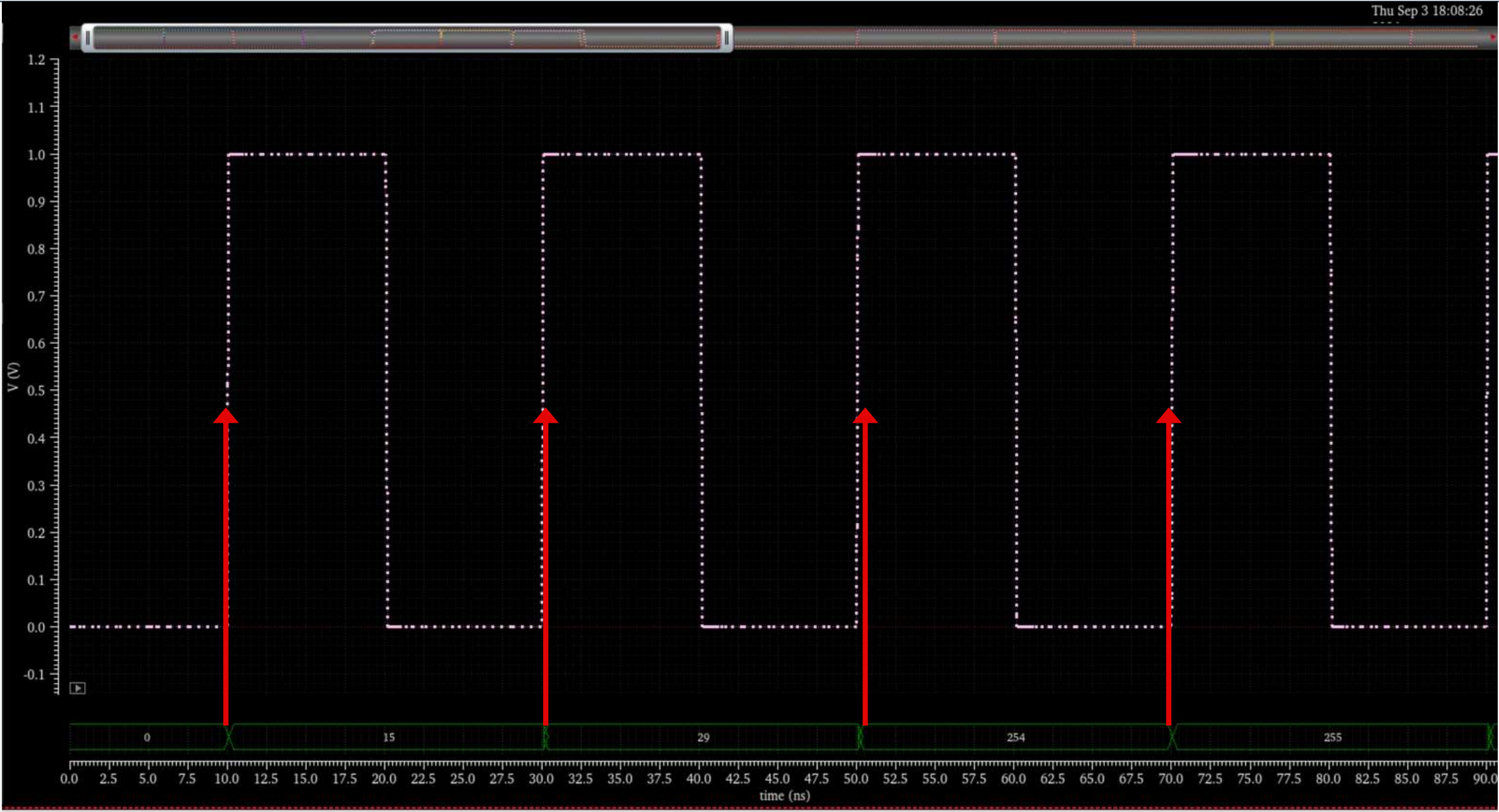


Simulation - Register

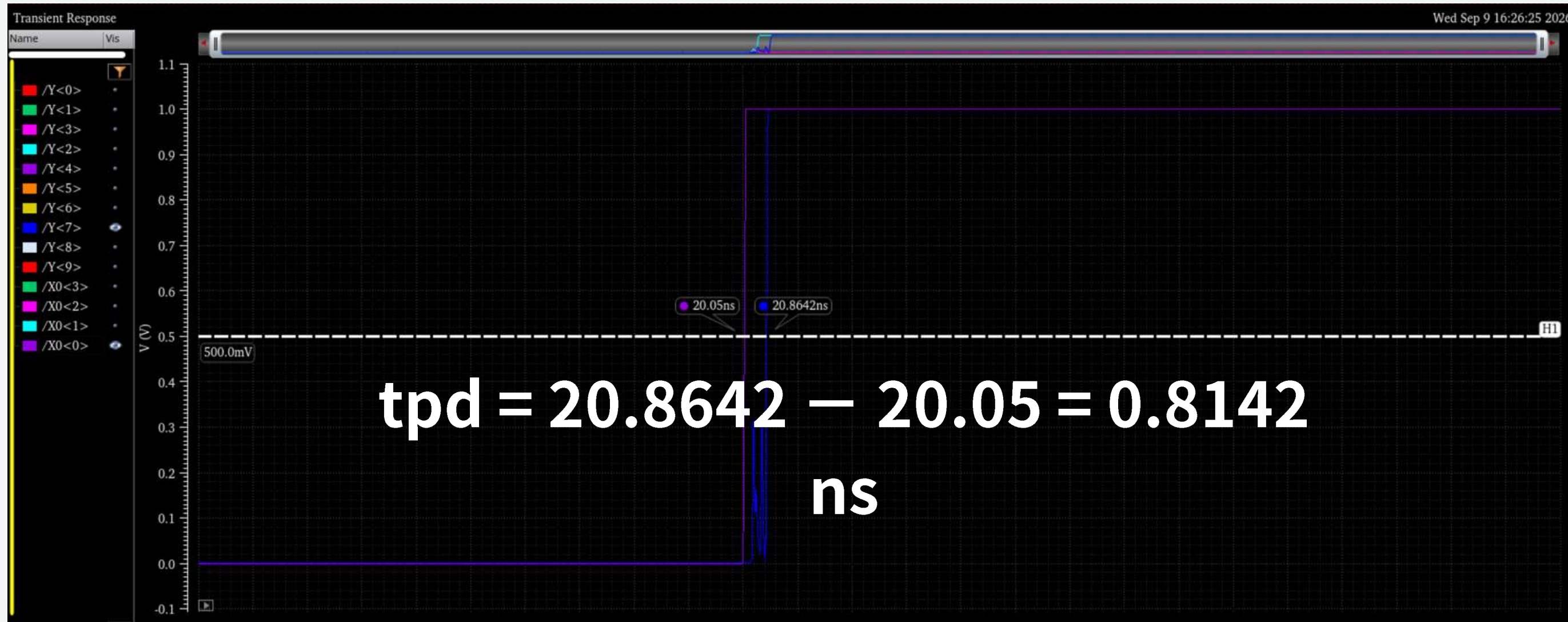
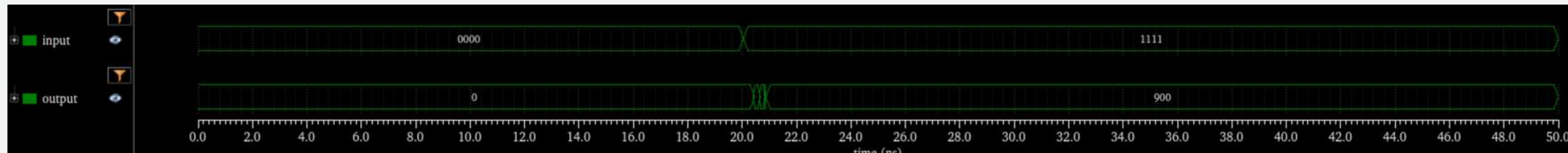


02 / Semiconductor Design

Simulation - MAC



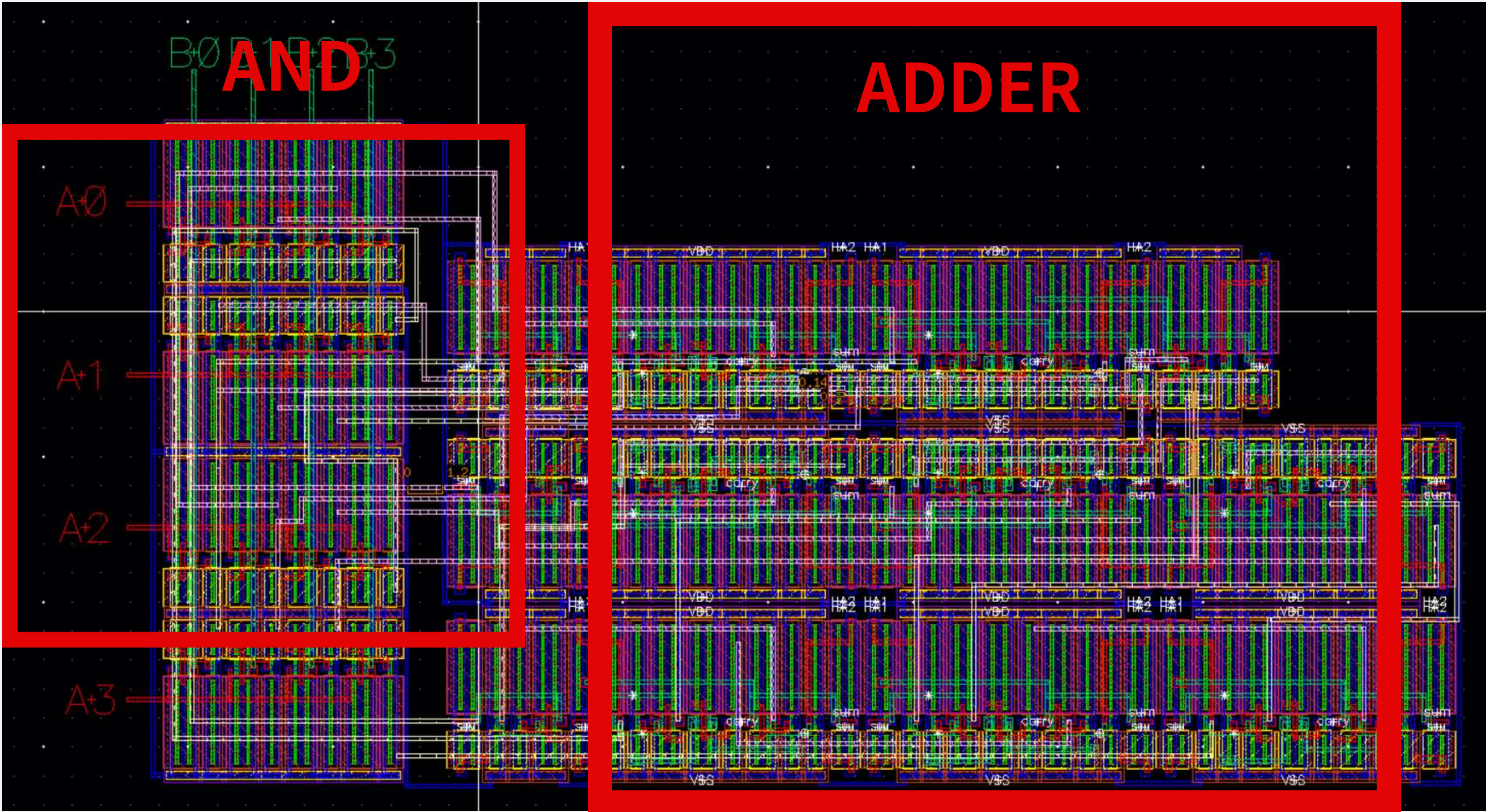
Simulation - Dot Product Accelerator



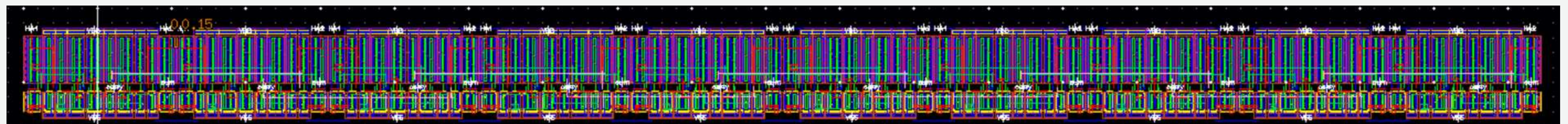
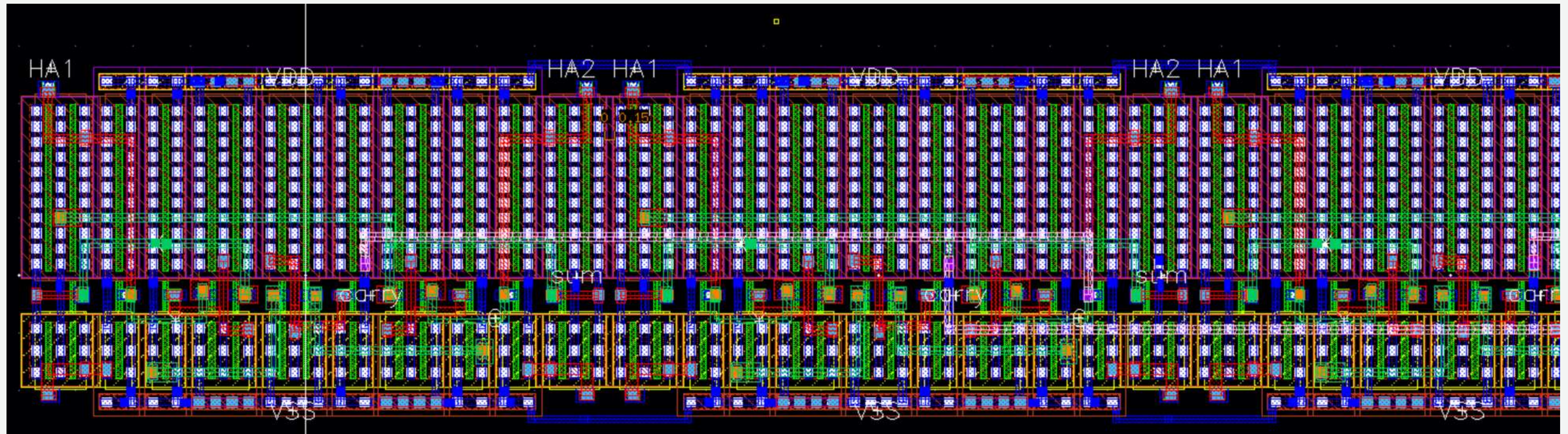
Layout - Metal

Metal Layer	주요 용도 / Routing Strategy
M1	Cell 내부 local connection, VDD/VSS rail, pin access
M2	AND input bus와 short/local signal
M3	M2와 교차하는 bus, vertical/inter-row connection
M4	AND → ADDER partial-product long routing, congestion relief
M5	복잡한 inter-block crossing 발생 시 reserve
M6	MAC/top-level global routing reserve
M7	최종 congestion 해결 또는 global net용 reserve

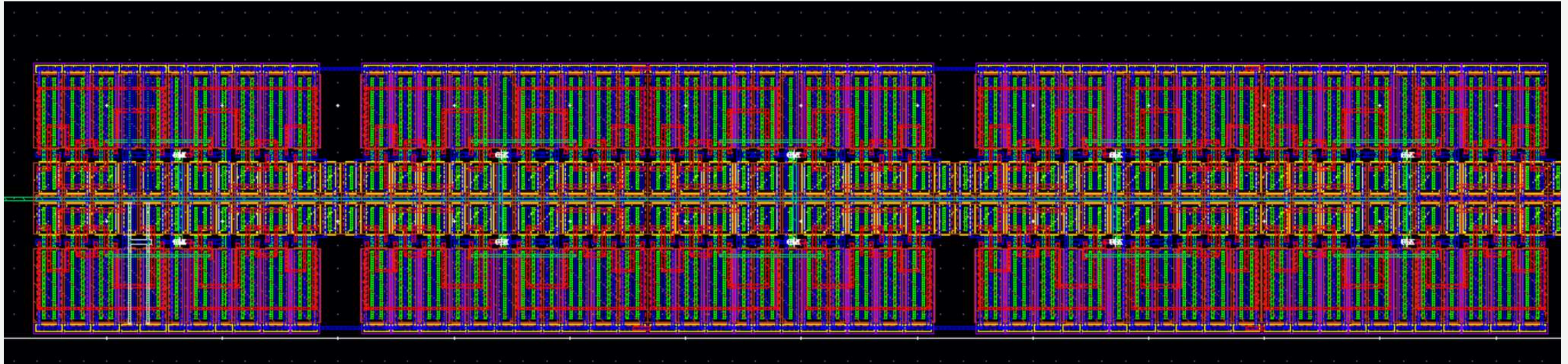
Layout - Multiplier



Layout - RCA10



Layout - Register

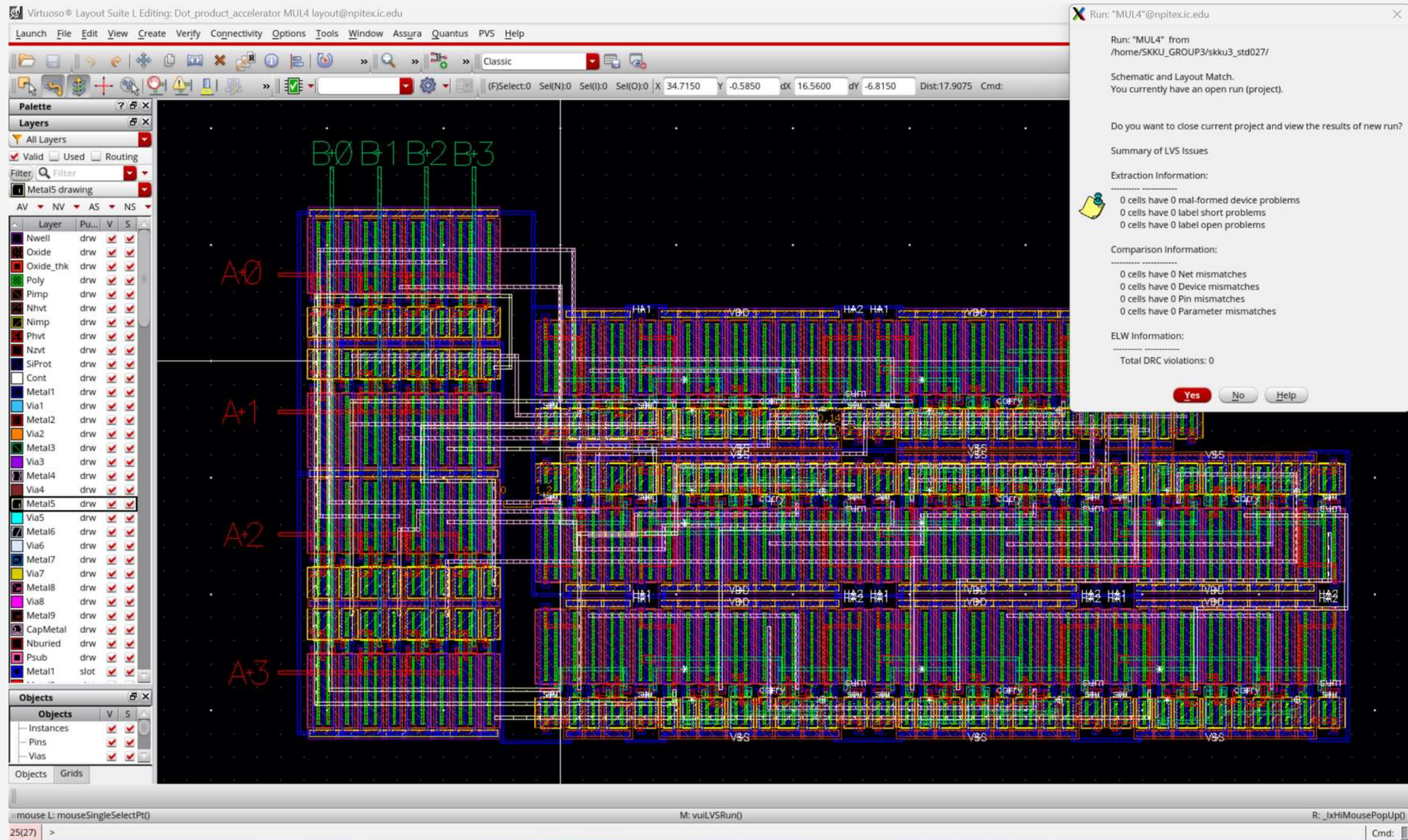


Layout - MAC

Layout - Dot Product Accelerator

02 / Semiconductor Design

DRC/LVS - Multiplier



DRC/LVS - Multiplier : Mapping Optimization

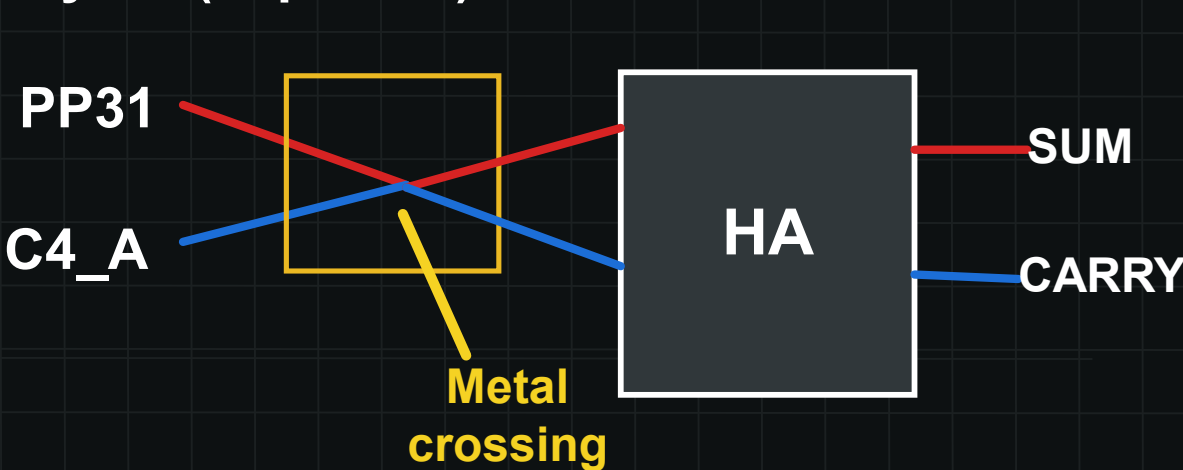
Before : Original Pin Mapping

Input routing crosses each other

Schematic (HA)



Layout (Top View)



- Input nets cross each other
- Longer routing length
- Higher metal congestion

Logical Equivalence

Half Adder (HA)

$$\begin{aligned} \text{SUM} &= A \oplus B \\ \text{CARRY} &= A \cdot B \end{aligned}$$

A and B are commutative
→ Input order can be swapped
without changing functionality

Full Adder (FA)

$$\begin{aligned} \text{SUM} &= X \oplus Y \oplus C_i \\ \text{CARRY} &= X \cdot Y + (X \oplus Y) \cdot C_i \end{aligned}$$

X and Y are commutative
→ Only X/Y can be swapped
(C_i remains unchanged)

Change input mapping
to match layout routing
and update schematic

After : Optimized Pin Mapping

Input routing is simplified (no crossing)

Schematic (HA)



Layout (Top View)



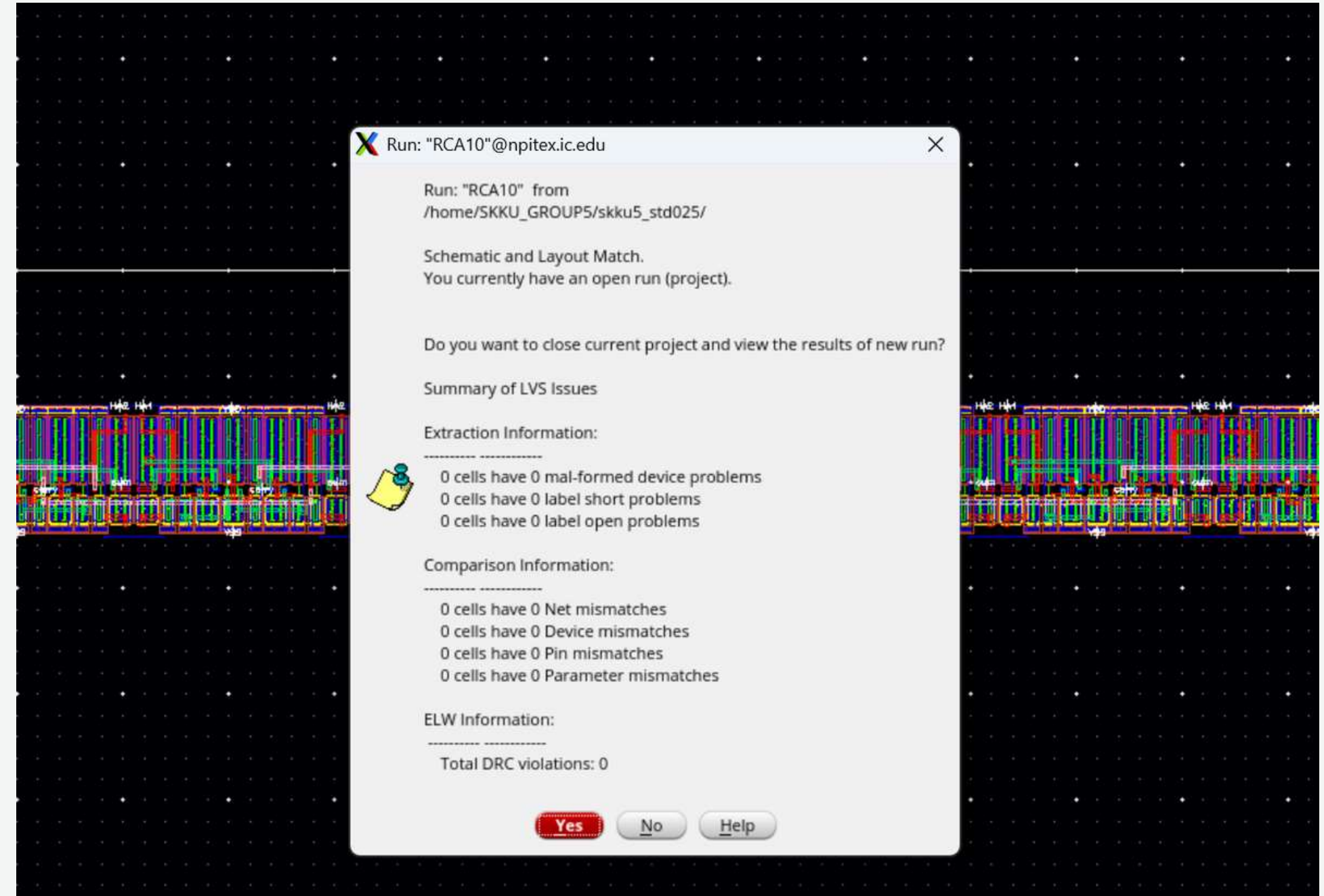
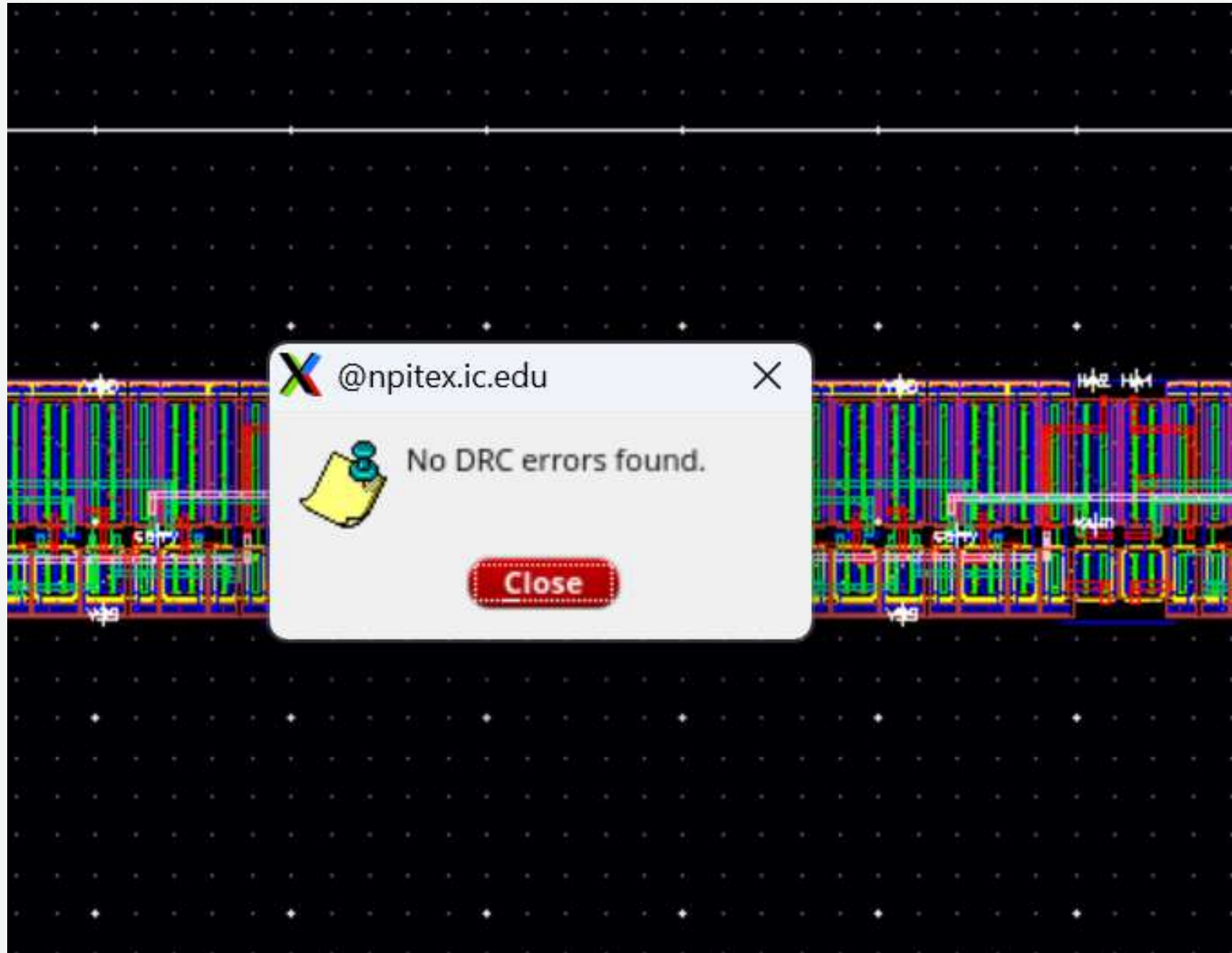
- No metal crossing
- Shorter routing length
- Lower metal congestion
- Same logical function (LVS verified)

Functional equivalence was preserved while optimizing physical connectivity.

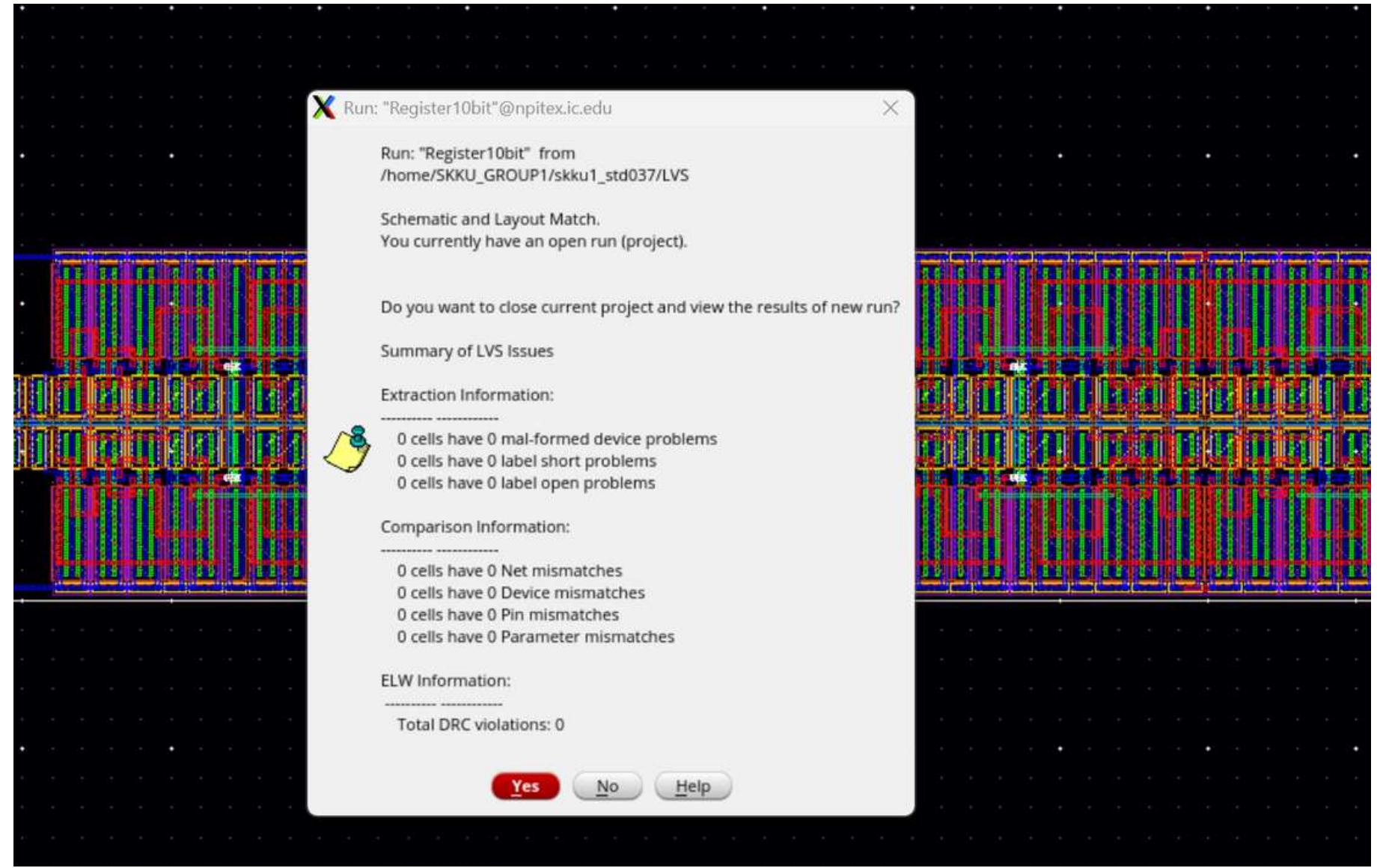
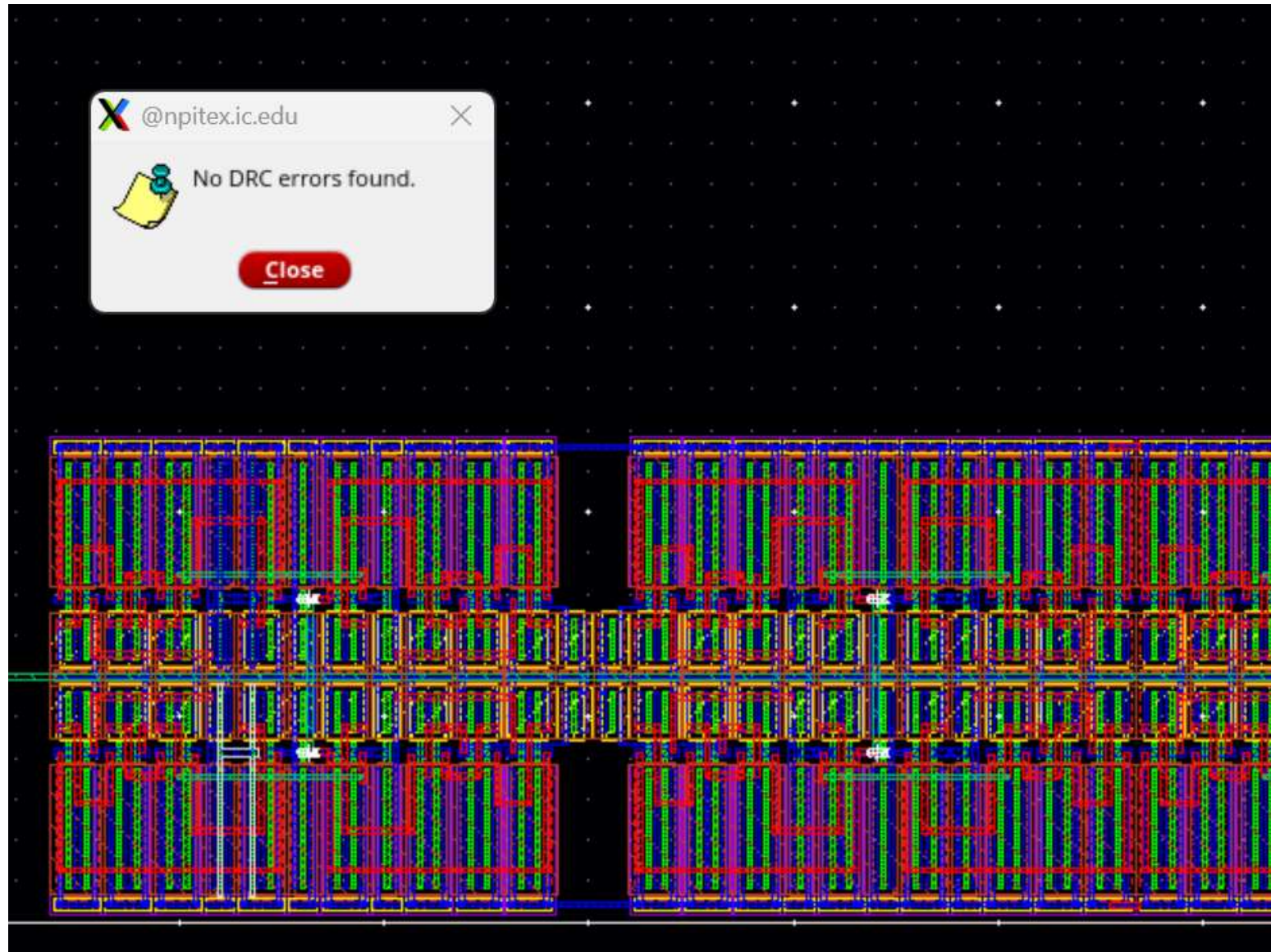


LVS Pass
Layout Optimized

DRC/LVS - RCA10



DRC/LVS - Register



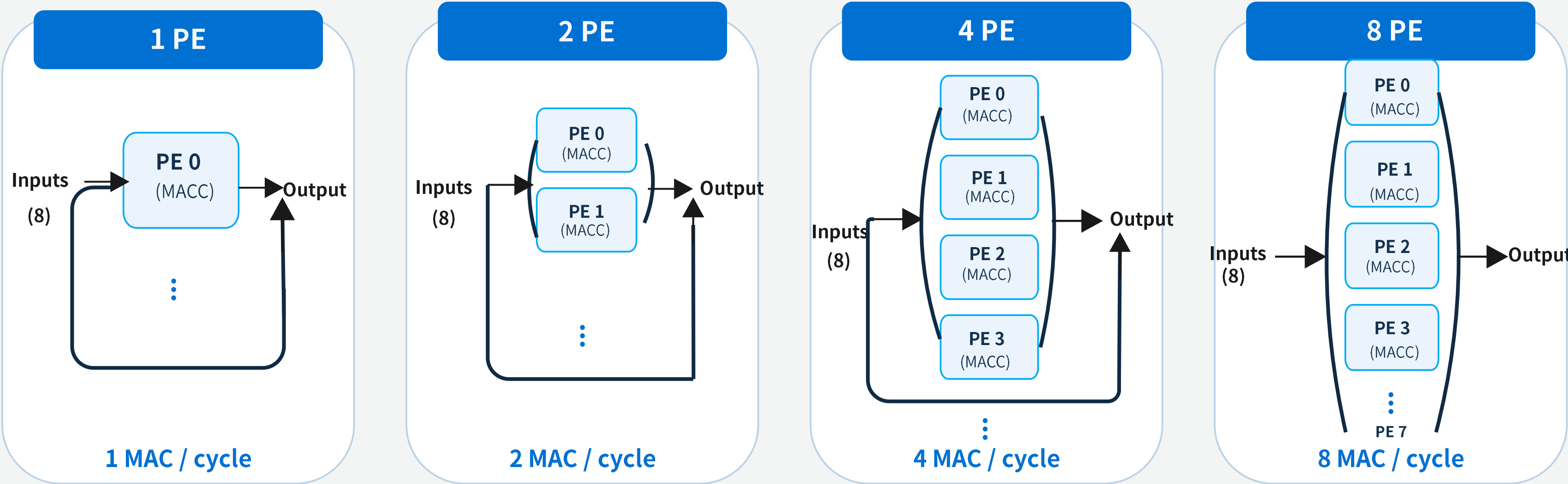
DRC/LVS - MAC

DRC/LVS - Dot Product Accelerator

03 / Conclusion

Scalability

8-term (N=8) Dot Product 기준, PE(=MACC) 수를 확장하며 성능 및 LAYOUT 분석



MAC operations / cycle	1	2	4	8
Cycles for 8-term Dot Product (N=8)	8 cycles	4 cycles	2 cycles	1 cycle
Ideal Speedup (vs. 1 PE)	1×	2×	4×	8×

→ PE 수 증가 → 처리량(Throughput) 향상 | Area / Power 증가 → Trade-off 분석 필요

Expected Results & Conclusion

Architecture	Advantage	Limitation
Sequential MAC	Small Area Low Power	Low Throughput
Parallel Dot Product Accelerator	High Throughput	Large Area / Power

Key Takeaways

- 01

Sequential 구조는 최소 area에 적합

Resource sharing으로 Hardware cost 최소화
- 02

Parallel 구조는 throughput과 latency에 유리

동시 연산량 증가로 처리 시간 단축
- 03

Post-layout PPA로 실제 trade-off를 검증

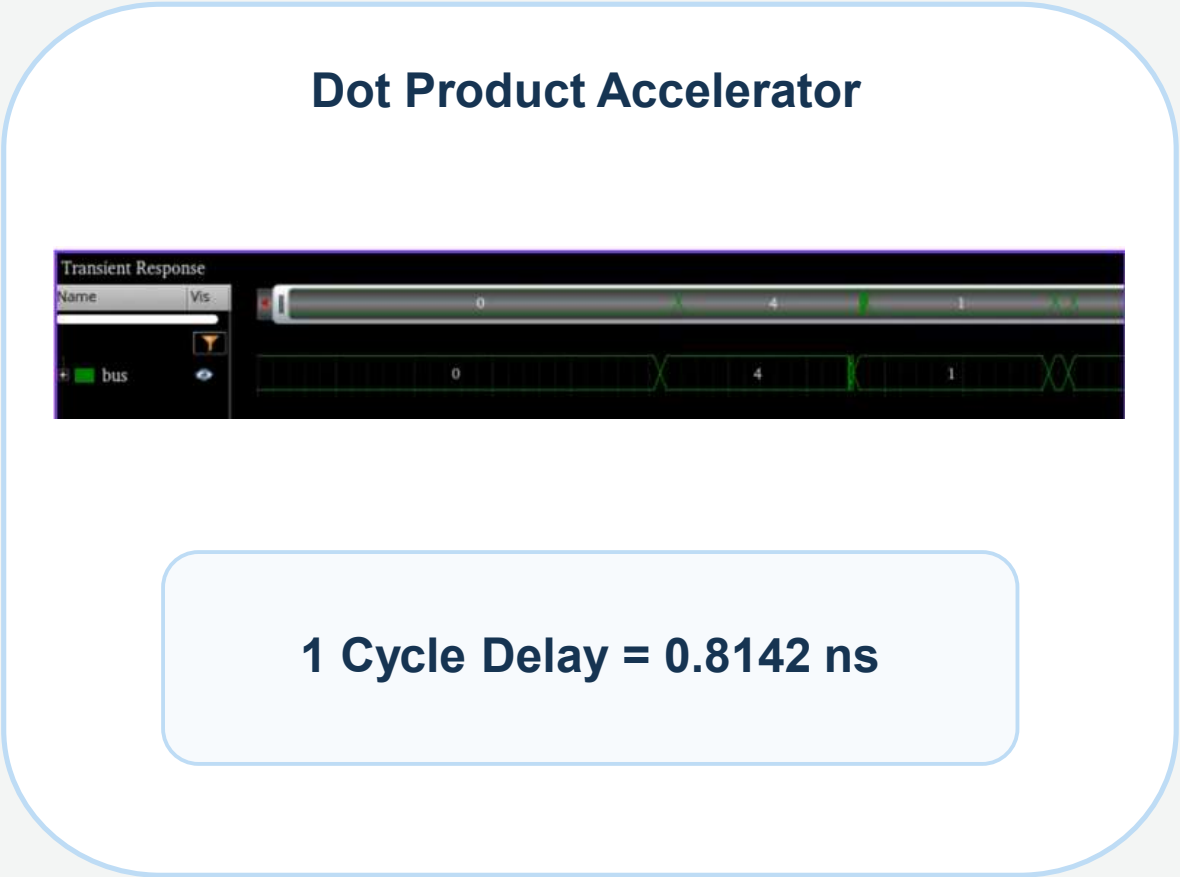
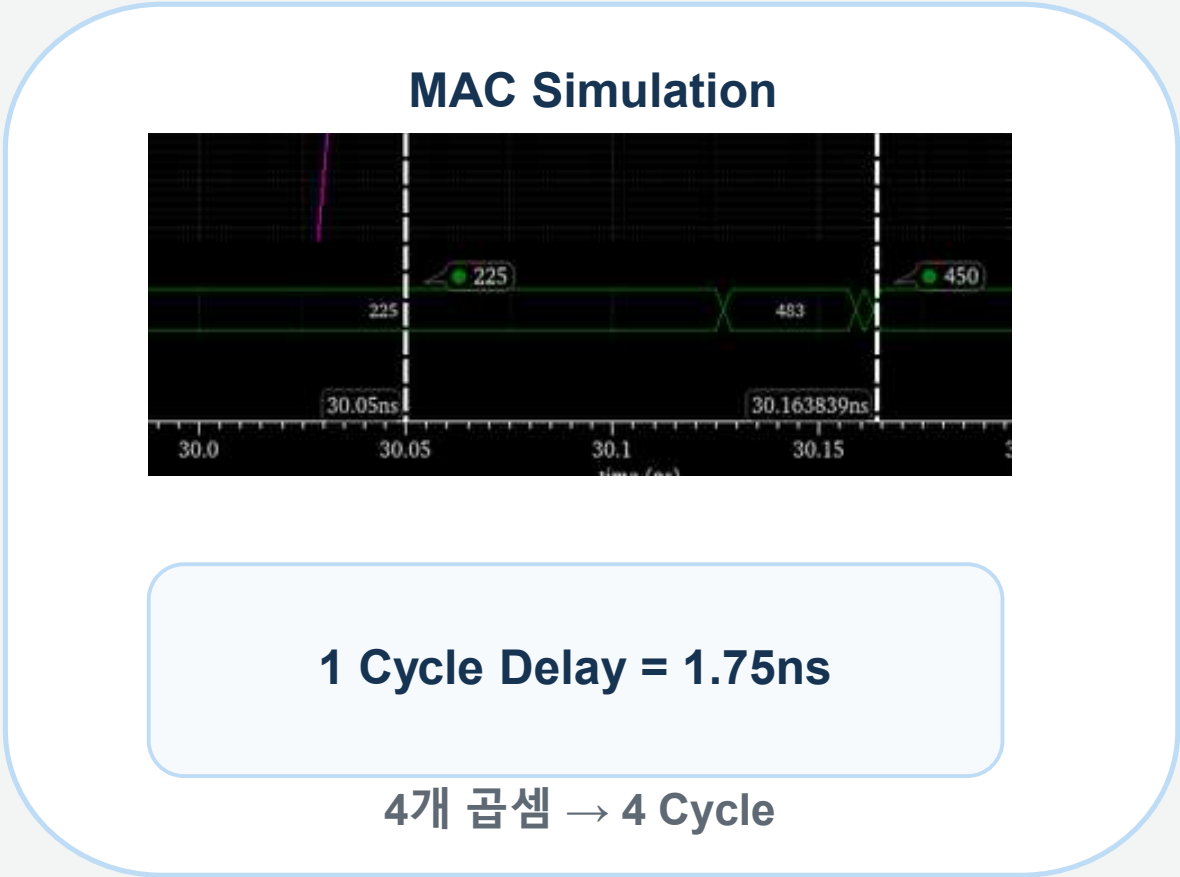
Area · Power · Delay · Throughput · Energy/Op

Conclusion

Parallelism increases throughput and reduces latency, while area/power overhead must be evaluated.

Actual Results & Conclusion

- Test Case(1111* 1111) 결과 분석

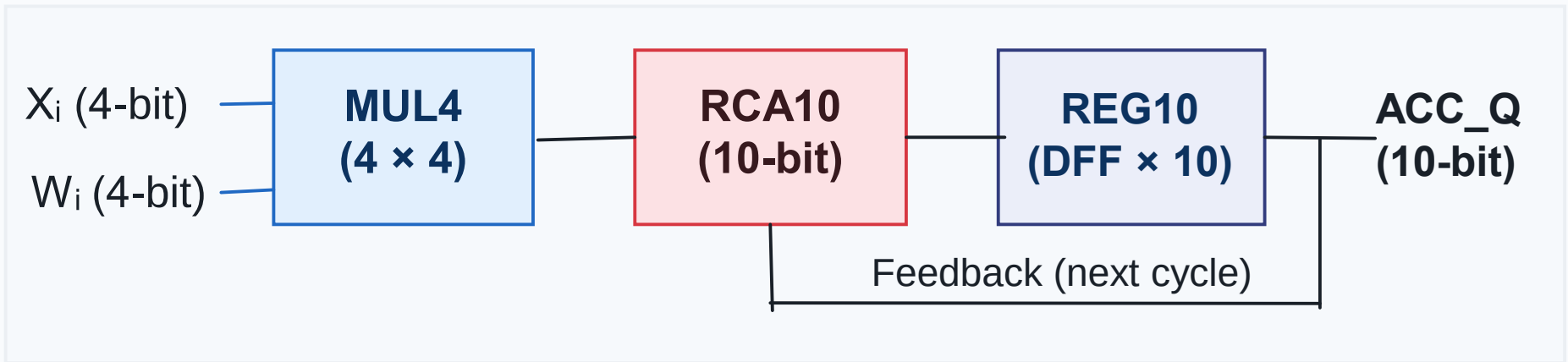


항목	MAC	Dot Product Accelerator
연산 구조	순차 연산	병렬 연산
1 Cycle 연산량	1 Multiplication	4 Multiplications
4개 곱셈 처리	4 Cycle	1 Cycle
1 Cycle Delay	1750 ps	814 ps
4개 연산 완료 시간	5330 ps	814 ps
Throughput	×1	×6.5

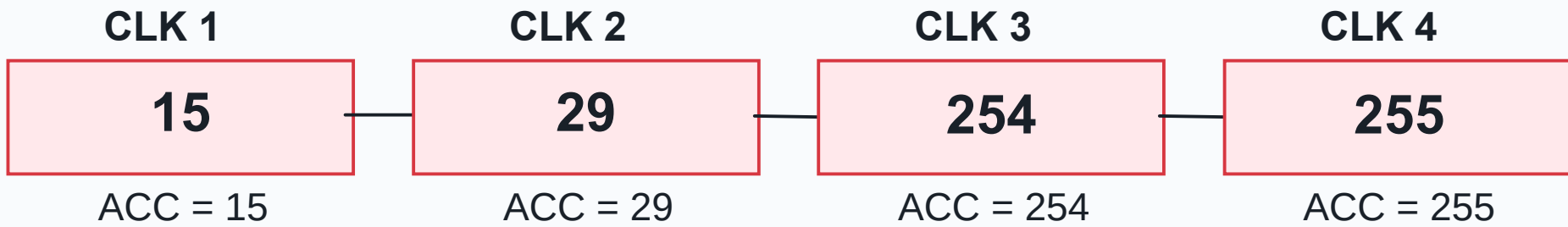
→ DOT Product Accelerator가 약 6.5배 빠름!

Actual Results & Conclusion

Sequential MAC (Single Multiplier, Accumulation with Feedback)



Operation (4-term example)

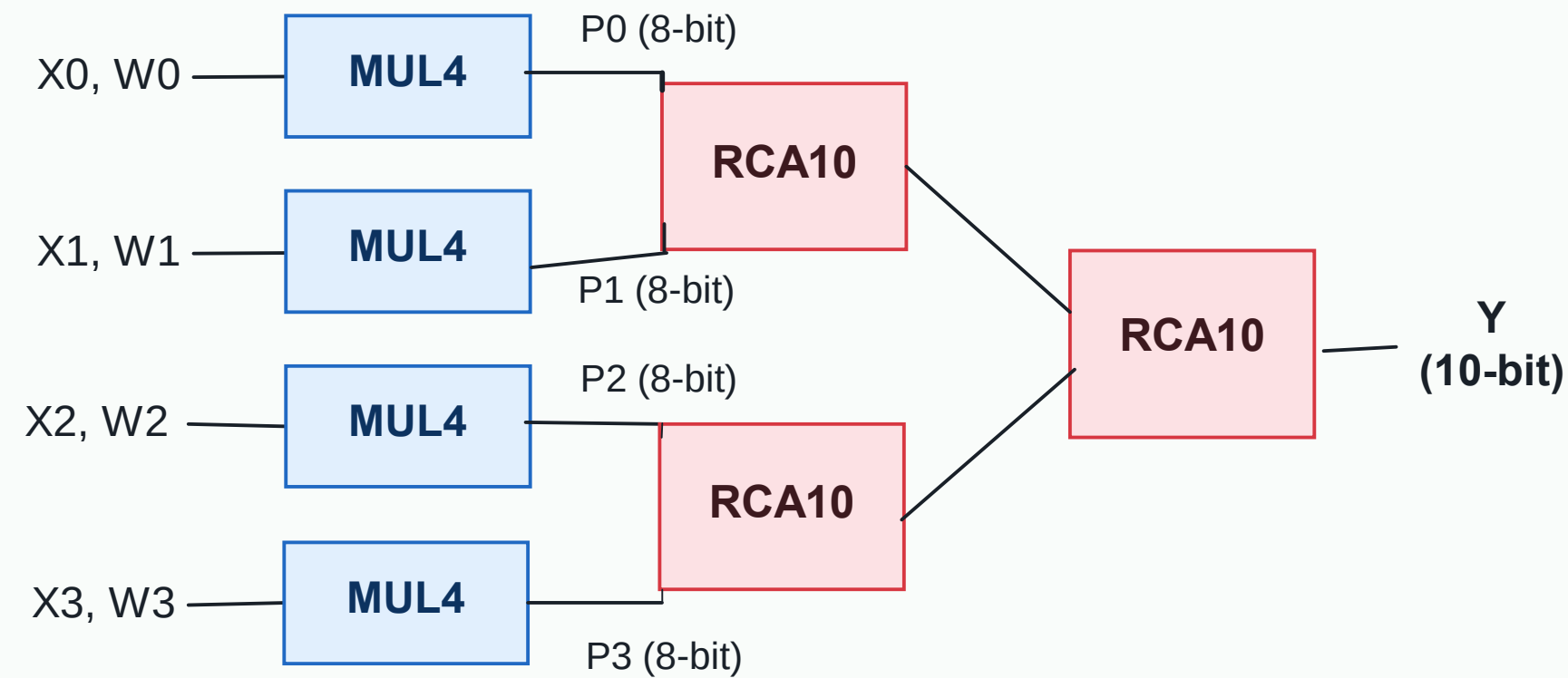


Each multiplication result is added in a different clock cycle because the updated value must be stored and fed back.

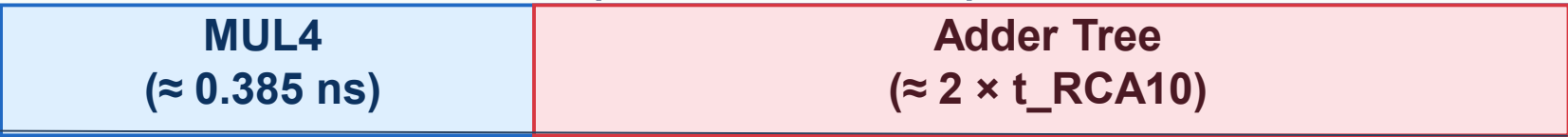
Clock period (minimum)
 $T_{CLK} \geq t_{MUL4} + t_{RCA10} + t_{setup}$
($t_{MUL4} \approx 0.385$ ns)

Latency for 4-term
 $\approx 4 \times T_{CLK}$

Parallel Dot Product Accelerator (Four Multipliers, 2-Stage Adder Tree)



One-shot computation (combinational)



$T_{DP} \approx t_{MUL4} + 2t_{RCA10}$

All four multiplications and additions are completed in a single evaluation. No feedback or extra clock cycle is required.



As long as sequential accumulation requires clocked feedback, the parallel dot product accelerator can compute all four products in a single evaluation, reducing latency at the cost of additional hardware.

Team Contribution

MEMBER	RESPONSIBILITY
전제현	RCA Schematic/Layout
곽대현	Multiplier Schematic/MAC Simulation
김규민	Dot Product Accelerator Schematic/Simulation
박재호	Multiplier Schematic/Layout
이준민	Register Schematic/Layout